

Program overview

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Year 2023/2024
 Organization Civil Engineering and Geosciences
 Education Master Applied Earth Sciences

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
AES Master 2023 (New Master Programme)							
AES Master Year 1							
AES Faculty Core Module (MUDE)							
CEGM1000	Modelling, Uncertainty and Data for Engineers	12					
AES Programme Core Modules							
AESM1001	Earth System, Natural Resources and Climate	7					
AESM1002	Dynamics of Solids and Fluids	5,5					
AESM2001	Physical Principles of Earth System Observation	5,5					
AES Discipline Climate & Weather							
AESM3001	Atmosphere Dynamics and Air-Sea Coupling	6					
AESM301A	Climate Physics	9					
AESM308A	Climate Modelling and Remote Sensing	9					
AESM309A	Climate Change and Dynamic Landforms	9					
AESM401B	Climate and Weather B-module	15					
AES Discipline Earth Observation							
AESM3002	Earth Observation Technologies	6					
AESM302A	Geo-Data Analysis and Geodesy	9					
AESM303A	Geo-Data and Geo-Informatics	9					
AESM307A	Earth Deformation Processes Across Scales	9					
AESM309A	Climate Change and Dynamic Landforms	9					
AESM402B	Earth Observation B-module	15					
AES Discipline Geo-Energy Engineering							
AESM3003	Geo-Energy Engineering Applications	6					
AESM304A	Flow and Simulation of Subsurface Processes	9					
AESM305A	Characterisation of the Subsurface	9					
AESM307A	Earth Deformation Processes Across Scales	9					
AESM309A	Climate Change and Dynamic Landforms	9					
AESM403B	Geo-Energy B-module	15					
AES Discipline Geo-Resources							
AESM3004	Economic and Structural Geology	6					
AESM306A	Extraction Processes and Consequences of Raw Materials	9					
AESM404B	Geo-resources B-module	15					
AES Master Year 2							
AES Electives							
AESM5110C	Aerosol and Cloud Microphysics	5					
AESM5120C	Data Assimilation for Geosciences	5					
AESM5210C	Climate Remote Sensing	5					
AESM5220C	Microwave Remote Sensing of the Earth's Surface	5					
AESM5230C	Coastal Remote Sensing	5					
AESM5240C	Applied Space Geodesy	5					
AESM5310C	Geo-Energy Integration Project	10					
AESM5320C	Outcrop Geology for Subsurface Characterization	5					
AESM5410C	Occupational Health and Safety Management	5					
AESM5420C	Geo-Resource Research Driven Project	15					
AES Cross Overs							
CEGM2000	Suspensions, Sludges and Soils	10					
CEGM2001	Sustainable Cities	10					
CEGM2002	Engineering for Global Development	10					
CEGM2003	Data Science and Artificial Intelligence for Engineers	10					
CEGM2004	Noise and Vibration: Generation, Propagation and Effect on Humans and Environment	10					
CEGM2005	Probabilistic Modelling of real-world phenomena through Observations and Elicitation (MORE)	10					
CEGM2006	Subsurface Storage for Energy, Water and Climate Applications	10					
CEGM2007	Resilient Deltas	10					
CEGM2008	Monitoring of Structural Health and Geohazards	10					
AES Thesis							
AESM6000	Thesis Preparation	5					
AESM7000	Master Thesis Applied Earth Sciences	30					

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AES Master Year 1

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AES Faculty Core Module (MUDE)

CEGM1000	Modelling, Uncertainty and Data for Engineers	12
Responsible Instructor	Dr.ir. R.C. Lanzafame	
Module Manager	Dr.ir. R.C. Lanzafame	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Instructor	F.P. van der Meer	
Instructor	D. Mehta	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. R. Taormina	
Instructor	A. Jagadeesh	
Instructor	Dr. P.M.N. Mares Nasarre	
Instructor	T.R. van Woudenberg	
Instructor	J.C. Pittman	
Contact Hours / Week x/x/x/x	4HC, 4W, 4PJ	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Modelling, Uncertainty and Data for Engineers (MUDE) is a module shared by students in all three programs within the Civil Engineering and Geosciences Faculty (Civil Engineering, Environmental Engineering, Applied Earth Sciences). In a typical week you will be introduced to new theory, practice it on your own through a series of exercises and work together with teachers and classmates in a medium-sized classroom setting. At the end of each week you will apply these concepts in a project along with a group of fellow students. You will also have weekly coding assignments to build your programming skills in a way that will help you apply the methods learned in MUDE using real data sets and models, as well as prepare you for your future MSC thesis work.	
Study Goals	After successfully completing the MUDE module a student is able to 1. Apply a diverse set of models and analysis techniques to solve engineering problems through a combination of mathematical formulation and implementation in computer code. 2. Process and analyze data from different sources and formats to gain insight on the underlying systems and incorporate in modeling approaches. 3. Mathematically formulate, solve and discuss estimation, risk, reliability and optimization problems for systems to support decision making. 4. Analyze and reflect on the methodology, performance, and results of modeling approaches and analysis techniques, accounting for uncertainty. 5. Implement and carry out analysis using Python programming language following specific coding standards and best practices while facilitating collaboration. 7. Communicate and interpret results of analyses with appropriate documentation, metrics and visualization techniques.	
Education Method	The weekly format is: - Mon: lecture - Tue: open question hours (attendance optional) - Wed: classroom workshops (teacher-led activities) - Fri: supervised group projects (bi-weekly assignments) Practice material is provided in the form of a digital textbook with analytic and programming exercises and is the primary focus of the Mon-Wed activities. These concepts are applied on Friday during the project sessions. Weekly coding assignments build fundamental programming skills that are also applied in the Friday project sessions.	
Assessment	- Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) The exams are paper-based. Project portfolio includes 7 bi-weekly projects (80%) as well as weekly coding assignments (20%). You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.	
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis, Python programming	
Academic Skills	Analysis, coding, communication, visualization, team skills	
Literature & Study Materials	An online digital textbook consisting of text, practice material and videos, as well as related materials (e.g., Jupyter Notebooks).	
Judgement	- Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) The exams are paper-based. Project portfolio includes 7 bi-weekly projects (80%) as well as periodic coding assignments (20%). You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.	
Permitted Materials during Exam	Calculator, pen/pencil. All formulas will be provided by the instructors.	
Collegerama	Yes No	

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AES Programme Core Modules

AESM1001	Earth System, Natural Resources and Climate	7
Responsible Instructor	R.J.G. Charton	
Module Manager	R.J.G. Charton	
Instructor	Dr. J.E.A. Storms	
Instructor	Dr. M.W.N. Buxton	
Instructor	Dr. M. Vizcaino	
Instructor	R.J.G. Charton	
Instructor	Dr. H.A. Abels	
Instructor	A. Melnyk	
Instructor	Dr. S.C. Toby	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	In this module, students will first be introduced to the Earth System and human interventions in the Earth System, including the exploitation of natural resources and greenhouse-gas emissions that caused climate change, and the societal challenge of moving towards a carbon-neutral world. Through three narratives, students will study through which processes and on what temporal and spatial scales the different Earth system components interact. The first story line is named Solid Earth & Resources and looks at the deep level dynamic processes that make the Earth a unique body in the solar system. These processes will be outlined in terms of the plate tectonic theory which provides a unified framework for the evolution of the solid Earth. We will examine how these processes have evolved through time and how they have been responsible for the distribution of the continents and the formation of mountain belts, volcanoes and the evolution of our natural resources. The second story line is named the Climate System, in which students will gain a basic understanding of the Earth's energy budget and the natural and anthropogenic influences on past (paleo-)climates and our current and future climate. They will be familiarized with the carbon cycle and study the role of the global atmospheric and oceanic circulation in setting climate zones and weather. The third story line is named Source to Sink and starts at mountains and follows the pathway of water and sediment towards the river, delta and coastal-oceanic basins. Along the way we will investigate formation of sediment, the water cycle, formation of stratigraphy, vegetation and land use changes, and how all this is affected by the past and current climate and weather. Finally, students will appraise and reflect on the societal and ethical implications of past or future human interventions related to resources and the climate of the Earth System.	
Study Goals	After completing this module, students will be able to: Analyse the different components of the Earth system and the processes and time and spatial scales on which the different components of the Earth system interact. [ILO A] Examine the processes that generate and deplete the availability of natural resources. [ILO A, D] Distinguish the processes that play a role in the Earth's energy (im)balance and calculate their impact on climate. [ILO A, C] Identify the processes that underlie impacts of anthropogenic activities on the Earth System. [ILO A, C] Reflect on social and ethical implications of human interventions in the Earth System. [ILO F]	
Education Method	Lectures and group practicals/exercises, feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Written exam, based on integrated understanding, and mixed group assignments (e.g., as a debate/poster/presentation) . Group assignments (30%): LOs 3, 5 Individual written exam (70%): LOs 1, 2, 3 and 4 Note: ethical reflection as part of assignment report (and optionally in exam as well)	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	none, although basic knowledge in geology will be useful	
Literature & Study Materials	Collaboration with peers, Integration of knowledge, reflection on ethics, report writing. Provided course material, published online lectures and material, book/e-books	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	none	

AESM1002	Dynamics of Solids and Fluids	5.5
Responsible Instructor	Dr. H. Hajibeygi	
Module Manager	Dr. H. Hajibeygi	
Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	A. Barnhoorn	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. A.M.H. Pluymakers	
Instructor	Dr. M.M. Boon	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides students with an understanding of basic concepts and mathematical solutions of fluid and solid dynamics with applications to atmosphere, oceans and the Solid Earth, preparing them for in-depth study of processes in specific Earth components. Students will learn how to derive the governing equations of fluid dynamics and solid dynamics, to explain how these are applied and simplified for flow and solid dynamics in the atmosphere, ocean and the solid Earth. Students will also carry out exercises as group assignments using python notebooks and model codes that exemplify simple models built on these governing equations. The topics covered are: Fluid dynamics: Governing Physical Equations Conservation of Momentum: equations of motion in an inertial frame and a rotating frame, Cartesian and spherical coordinates, advection, Eulerian and Lagrangian derivatives, vorticity equation, vorticity and circulation theorems, vorticity equation in a rotating frame Dimensional analysis and dimension-less numbers (Reynolds, Rossby, Prandtl, Froude, Peclet) Conservation of Mass: compressible and incompressible flows Equation of state, thermodynamic equations for ideal gases and for liquids and mixtures (first and second law of thermodynamics, Daltons law, phase behaviour). Buoyancy and thermal (static) stability Laminar versus turbulent flows and diffusion (as proxy of turbulence) Fluid dynamics: Applications to ocean and atmosphere Examples of flow in the atmosphere and oceans: geostrophic flows, convection, turbulence Boussinesq and anelastic approximation, vertical (pressure) coordinates, Hydrostatic balance. Primitive equations as basis of atmosphere and ocean circulation models. Fluid dynamics: Application to Flow in Porous Media Single-phase flow in porous media (Porosity, Permeability, Darcys law, Gov eq., solving potential, pressure & velocity) Single-phase heat-mass transfer in porous media (conduction and convection) Upscaling (heterogeneities, anisotropic) Diffusion/dispersion in porous media Introduction to multiphase flow in porous media (gov. equations, basics of kr & pc); Rock-fluid interaction. Solid Dynamics Solid Mechanics: elasticity, material strength, brittle to ductile deformation, effect of conditions (temperature, pressure, material properties). Applications to rock deformation in shallow and deep Earth, and ice flow.	
Study Goals	After completing this module, students will be able to: Distinguish flows and deformation in different parts of the Earth system (atmosphere, ocean and solid Earth). [ILO C] Perform dimensional analysis of flows in different media and explain appropriate approximations to the equation of motion. [ILO A] Characterise and derive the physical equations that underlie advection, diffusion, convection, conduction, elasticity, brittle and ductile deformation, and consider their application in models of Earth System components. [ILO A, C] Apply the physical equations to examples of flow and deformation in different media using python notebooks. [ILO C, D] Analyse and compare the behaviour of flow and deformation phenomena in the Earth System using simple python models. [ILO A]	

Education Method	Lectures and group practicals/exercises Feedback via quizzes (formative) and intermediate assignment results (formative).
Assessment	Assignments (50%) and a final exam (50%) Written exam is individual, Assignment is in a group.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Bachelor of Science in Engineering and Sciences, with basic knowledge of mathematics and physics.
Literature & Study Materials	1- Critical and algorithmic thinking 2- Design an algorithm 3- Writing scientific report 4- Presentation skills
Judgement	A mix of book chapters and lecture notes.
Permitted Materials during Exam	Reports and Exam.
Collegerama	NONE.
	No

AESM2001	Physical Principles of Earth System Observation	5.5
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Module Manager	Dr.ir. F. Lopez Dekker	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Dr. S.C. Steele-Dunne	
Instructor	Dr. C. Weemstra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Measuring is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Electromagnetic, seismic and gravity potential-field observations inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. The measurements can be acquired from spaceborne, airborne, surface and sub-surface-based sensors. This programme core module aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain geophysical variables. For example, students will learn how electromagnetic theory allows to use the intensity of radar echoes to yield information about rain rate, soil moisture, ocean roughness, or the layering of the subsurface. Similarly, they will learn how potential field theory can be applied to quantify mass changes of, e.g., the ice sheets. Students will be able to weigh the advantages and disadvantages of, for example, microwave versus visible and near-infrared observations for monitoring the Earth surface, and seismic and electromagnetic imaging in mapping the subsurface. Review basics of electromagnetic (EM) waves (propagation, polarization, spectra, Doppler, etc.) and propagation effects (attenuation, refraction, dispersion, polarimetric effects) Optical-infrared scattering and propagation (different reflection types, albedo, effect of wavelength, BRDF, Radiative transfer modelling of optical EM waves). Physical principles and limitations of VNIR/SWIR sensing systems (photographic systems, LiDAR, spectral systems including TIR) in terms of spatial and spectral resolution, sensitivity, atmospheric disturbances and corrections. Microwave scattering and propagation (Rayleigh, Mie scattering, rough surface, volume scattering, clear air, radiative transfer modelling of microwave scattering in the atmosphere and at the land surface). Radiometry (Radiometric Quantities and Units, Blackbody Radiation, emissivity at different frequencies). Radiative transfer modelling in passive microwave remote sensing of the Earths atmosphere and surface. Physical principles and limitations of microwave sensing systems (radiometer, non-imaging radar, imaging radar, ground-penetrating radar) in terms of spatial and spectral resolution, sensitivity, atmospheric propagation effects and corrections, radar range equation Seismic (including acoustic) waves (Newtons second law, stress tensor, equation of motion, elasto-dynamic wave equation, Plane P- and S-waves, spatial and temporal resolution) Potential fields (gravity, magnetic and electric fields) and their relationship to physical properties of a medium (Gauss law and its applications, Newtonian/Magnetic Potential, Laplace equation and harmonic functions, representation of potential fields, spherical harmonics) Choose an approach for the observation and measurement of a geophysical variable.	
Study Goals	After completing this module, students will be able to: Characterize electromagnetic waves and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and & subsurface). [ILO A] Describe the interaction of optical-infrared waves with the Earths atmosphere and surface and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of optical/infrared remote sensing in determining geophysical variables. [ILO A, D] Describe microwave scattering in the Earths atmosphere and at the Earths surface, and analyse the fundamental sensitivity to geophysical variables of interest. [ILO A] Describe the emission of thermal waves and microwaves from the Earths atmosphere and surface, and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of microwave remote sensing in determining geophysical variables. [ILO A, D] Explain mathematically how the elastodynamic wave equation is derived, characterize elastic waves, and explain their sensitivity to contrast parameters. [ILO A] Establish the linearized relationship between an observable and a given representation of the potential field, and synthesize the potential from functionals given on the surface of a sphere. [ILO A] Critically discuss the potential value and limitations of different types of measurements (I.e., different bands of the EM spectrum and EMR/seismic/potential field) and propose and defend a type of measurement to observe certain geophysical variables. [ILO F, G]	
Education Method	Lectures and group practicals/exercises. Feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Module is summatively assessed with a project, using 1 rubric (similar as thesis project) at the end of the period based on:	
	Assignment report (may comprise different parts, i.e., based on a several assignments, report includes reflection on LOs)	
	Presentation	
	Oral discussion (may include questions on LOs not directly covered in the assignment work).	

Expected prior Knowledge Literature & Study Materials Judgement	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Some background in electromagnetic theory and fundamentals of wave propagation. Solving linear partial and ordinary differential equations. Provided course material, published online lectures and material, book/e-books For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
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Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Discipline Climate & Weather

AESM3001	Atmosphere Dynamics and Air-Sea Coupling	6
Responsible Instructor	Dr. A.A. Nuijens	
Module Manager	Dr. A.A. Nuijens	
Instructor	Prof.dr. J.D. Pietrzak	
Instructor	Prof.dr. C.A. Katsman	
Contact Hours / Week x/x/x/x	0/0/16/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This discipline core will provide students with the deeper physical understanding to comprehend and study the complex atmospheric flows that define climate and weather. The module will use interactive lectures to introduce advanced dynamical principles to explain global climate, the planetary circulation, weather systems in the tropics and extratropics and coupled atmosphere-ocean phenomena that set weather and climate variability. In these lectures, laboratory experiments with a rotating tank or visualizations of atmospheric flows are a starting point to describe the complex flows in our atmosphere. Each week, the students will be guided during practical lab work/assignments in which material collected during the tank experiments, toy models or observational datasets are used and analyzed to describe uncoupled and coupled phenomena in the atmosphere.	
Study Goals	After completing this module, students will be able to: 1. Analyse data of the horizontal and vertical structure of atmospheric pressure, temperature, humidity and wind to explain the general circulation of the atmosphere and climate zones [ILO A,B] 2. Apply conservation of heat, moisture and momentum to explain the origin and features of tropical and extratropical weather phenomena [ILO A] 3. Assess the role of air-sea coupling on atmospheric dynamics on short (weather) and long (climate) timescales [ILO A] 4. Analyse and evaluate the sensitivity of circulations and climate by running experiments with idealized simulations and toy models [ILO A,B,C] 5. Design a hypothetical lab experiment to study the influence of climate change on atmosphere dynamics [ILO A,C,E] 6. Discuss processes that cause uncertainty in weather and climate prediction [ILO C,F,G] (ILO = Intended Learning Outcomes of the AES programme)	
Education Method	Plenary discussions and quizzes during lectures and labs, practical (lab) demonstrations with rotating weather tank, optional self-study assignments, midterm exam.	
Assessment	Written theory exam.	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	Faculty programme core, AES programme core.	
Literature & Study Materials	Demonstrating conceptual understanding through the application of theory to new situations using relevant equations. Extracting, evaluating and discussing modeling output and observational data. Creating new experiments.	
Judgement	Textbook: Marshall, Plumb: Atmosphere, Ocean and Climate Dynamics: An Introductory Text. Lecture notes provided with slides.	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	Lecture notes, personal notes and calculator (open book)	
	No	

AESM301A	Climate Physics	9
Responsible Instructor	Dr. A.A. Nuijens	
Module Manager	Dr. A.A. Nuijens	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. A.P. Siebesma	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. M. Vizcaino	
Instructor	Dr.ir. B.J.H. van de Wiel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This module is aimed at students with an interest in the field of applied meteorology such as wind and solar energy prediction or climate impacts on land use), and students interested in the design, analysis and assessment of weather and climate models. The main objective of the module is to provide students with a solid understanding of atmospheric processes that play a key role in climate and that have a strong impact on daily life: winds, clouds and radiation, storms and precipitation. This module will explain students how these small-scale processes are coupled to large scale circulations and why their formulation in general circulation and climate models is particularly important for climate prediction. The module comprises two units that focus on deepening process knowledge and analytical thinking: Land-atmosphere interactions and Water in the atmosphere. These will cover atmospheric processes strongly linked to turbulence and radiative transfer across all scales that are not explicitly represented in general circulation models. The third unit called Climate Modelling provides students with hands-on experience in the use and application of state-of-the-art general circulation models, including an assessment of the uncertainties in current climate projections.	
Study Goals	After completing this module, students will be able to: 1. Explain the physics and dynamics of the transport of energy, water and momentum between the surface, the atmospheric boundary layer and the broader atmospheric circulation [ILO A] 2. Apply simplified models of turbulence and energy and water exchange, including convection and clouds [ILO B,C] 3. Reflect on the parameterization and coupling of atmospheric processes in general circulation models and their role in current uncertainties in climate prediction [ILO C,F,G] 4. Analyse climate simulations for the purpose of process understanding as well as assessment of climate change impacts and mitigation/adaptation policies [ILO A,B,C,E,F] 5. Assess the influence of land-atmosphere coupling, turbulence, convection and clouds on large-scale circulations (weather) and climate. [ILO A,C,F] 6. Hypothesize how land-atmosphere coupling, turbulence and convection and clouds are influenced by changes in weather and climate [ILO A,C,F] (ILO = Intended Learning Outcomes of the AES programme)	
Education Method	Unit Water in the Atmosphere: - 2 hours of classical lectures and 2 hours of supervised computer-based and regular exercises. Unit Land-Atmosphere Interaction: - 2 hours of lectures including short mini-exercises, table-top experimental demonstrations, article discussion in class, 2 hours of supervised theoretical exercises. Unit Climate Modelling: - Lectures, exercises (including paper analysis/discussion), guided assignments See also Brightspace for updated and more information of this module's units.	
Assessment	Summative assessment: Two written exams on unit level (25% each), 1 assignment on unit level (25%) and one module level synthesis assignment (25%). Formative assessment: Unit Water in the Atmosphere: - Computer-based and regular (homework) exercises are supervised with ample feedback. Unit Land-Atmosphere Interactions: - Exercises made under supervision of the instructors, feedback to the experimental work by the students (both on performance and on reporting) and a midterm exam. Unit Climate Modelling: Exercises (including paper analysis/discussion) and assignments	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	Combination with DC-CW (Atmosphere Dynamics and Air-Sea Coupling).	
Literature & Study Materials	Deepening scientific knowledge, analytical and critical thinking, problem solving, synthesizing information and scientific writing. Unit Water in the Atmosphere: - "Atmospheric Science, An Introductory Survey", by John Wallace and Peter Hobbs Unit Land-Atmosphere interaction: - The book: Transport in the Atmosphere-Vegetation-Soil Continuum, by Moene and Van Dam, Cambridge University Press. Additional background material and/or papers will be provided via BrightSpace. Experimental material is provided in the class.	

	Unit Climate Modelling: - Primary Course Book: Climate Modeling Primer (second course book and Practice) by McGuffie and Henderson-Sellers. - Secondary Course Book: Climate System Modeling, by Kevin Trenberth - IPCC reports (Practice) https://www.ipcc.ch/reports/ See Brightspace for more detailed information on literature and study materials for the module and units.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Calculator. See Brightspace for updated information on permitted material during the exam.
Collegerama	No

AESM308A	Climate Modelling and Remote Sensing	9
Responsible Instructor	Dr. S.C. Steele-Dunne	
Module Manager	Dr. S.C. Steele-Dunne	
Instructor	Prof.dr.ir. R.F. Hanssen	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkooei	
Instructor	Dr.ir. D.C. Slobbe	
Instructor	Dr. M. Vizcaino	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module complements the Climate & Weather core module. It is aimed at students from the Climate & Weather learning line who are interested in the synergy offered by the combination of remote sensing and climate models to understand, predict and model the wide range of processes governing our present-day coupled climate system. The objective of this module is to provide students with a physical understanding of these processes and the skills needed to observe and assess those processes using ground-based and space-borne observations, and coupled global models. Ultimately, the student will be able to understand and analyse both one specific component of the climate system (e.g. cryosphere, terrestrial water cycle, ocean), and on a broader level, the interactions between them using a combination of observations and modelling. Students will learn how to formulate, consolidate, and prioritize their data needs, and will draw on their prior/existing understanding of the underlying physical principles to select a suitable observation technique to observe the variable(s) of interest. By developing a preliminary design for an observational mission/campaign they will become familiar with the trade-offs commonly made to reconcile requirements from a modelers perspective with the capabilities of the selected techniques. They will analyse real or synthetic observation data, perform a quality assessment, estimate the parameters of interest and reflect on the ability of the mission/campaign to meet their needs in terms of parameterizing processes of interest, constraining models and improving predictions. The module comprises two units that, together, offer the students a thorough understanding of how large-scale climate processes can be observed and modelled. Earth Observation Technologies (6 EC) consists of the core module of the Earth Observation learning line and familiarizes students with the process of specifying and designing an observational mission/campaign (e.g., a new satellite mission or a ground measurement campaign). This process includes the interpretation and analysis of user needs and their translation to observational requirements, the high-level design of possible technical solutions, and the evaluation of the expected observational performance with respect to user requirements. The unit Climate Modeling and Prediction (3 EC) from the Climate & Weather learning line is shared with other modules and provides students with hands-on experience in the use and application of state-of-the-art general circulation models, including an assessment of the uncertainties in current climate projections.	
Study Goals	After completing this module, students will be able to: Formulate, consolidate, and prioritize remote sensing data requirements to study processes in the coupled climate system and validate and inform their parametrization in climate models [ILO A,E,F] Select a suitable observation technique to observe the variable(s) of interest, drawing on their prior/existing understanding of the underlying physical principles [ILO A,E] Characterize real or synthetic observation data, perform a quality assessment, estimate the parameters of interest and reflect on the ability of the available data to meet their requirements. [ILO A,B,C,F,G] Construct (simplified) models of climate variables and evaluate these with remote sensing data. [ILO A,B,C,E] Analyse climate simulations for the purpose of process understanding (climate science) as well as for the interpretation of remote sensing data records. [ILO A,B,C] Reflect on how integration of remote sensing data in climate modelling may improve our understanding and prediction of climate variables. [ILO A,C,E,G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	unit 1: project unit level (67%, 1-dec, 5.8, compulsory); Unit 2: Assignment (25%, 1-dec, 5.8, compulsory); and one module level synthesis assignment (8%, 1-dec, 5.8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Academic Skills	Combination with DC-CW.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	See Brightspace for more detailed information	
Colleggerama	No	

AESM309A	Climate Change and Dynamic Landforms	9
Responsible Instructor	Dr. M. Vizcaino	
Module Manager	Dr. M. Vizcaino	
Instructor	Dr.ir. H. van der Marel	
Instructor	Dr. J.E.A. Storms	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. R.C. Lindenberg	
Instructor	Dr. H.A. Abels	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module is aimed at students with an interest in the physics of climate change, its effects on the natural environment and how those can be analysed through geospatial data. In the climate modelling unit, student will learn about the numerical, computational and modelling concepts that underlie general circulation models and coupled climate models, which are the primary tools to explain and/or predict the dynamics of past and future climate. The natural environment is here represented by rivers and deltas, which provide an instructive and societal-relevant study case to learn about the interaction between climate change and the surface of the solid Earth. Finally, the Earth observation unit has a strong hands-on identity, which complements the geological characterization of landforms by teaching how to deliver remote sensing products that can be used in geophysical studies. This module will fit into a study-route that addresses how the natural environment is affected by past and future climate change at decadal to secular scales, by using numerical models of the driving physical processes as well as geological and instrumental observations.	
Study Goals	After completing this module, students will be able to: Analyse the interaction between climate change and the solid Earth's surface, with particular attention to the timescales involved. [ILO A,C] Explain the origin of present-day morphodynamic features by integrating geological and geospatial data. [ILO A,B] Analyse climate simulations for the purpose of understanding the driving physical processes. [ILO C] Analyse source-to-sink sedimentary systems and records in relation to tectonics and climate amongst other parameters. [ILO A] Extract geophysical parameters from geospatial data. [ILO B] Present findings in a precise and organized way, both numerically as well as graphically [ILO A,G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Unit 1-3 assignment on unit level (25% each, 1-dec, 5,8, compulsory) and one module level synthesis assignments (25%, 1-dec, 5,8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM401B	Climate and Weather B-module	15
Responsible Instructor	Dr. M.A. Schleiss	
Module Manager	Dr. M.A. Schleiss	
Instructor	C.M.H. Unal	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr.ir. B. Wouters	
Instructor	Dr. M. Vizcaino	
Instructor	Dr. D.P. Donovan	
Instructor	Dr. H. Zekollari	
Instructor	R.I. Mackenzie	
Contact Hours / Week x/x/x/x	week 1: 12 PJ week 3, 5, 6 : 8 PJ week 7, 8: PJ	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The lab and the theory part of the B-module are connected and inquiry-based. Theory (8EC) Students can choose theory units depending on the choice for a Climate Impacts or Geohazards Lab. Climate data analysis is compulsory. Two units are to be selected from subset 1, and one unit is to be selected from subset 2. Lab (4EC) Individual or groups of max. 3 students The lab has a central place in the module, triggering the inquiries and as the place where the students apply the theory to a challenge related to monitoring and/or prediction of climate impacts or geohazards. The student (team) defines a topic related to the challenge within a predefined theme that they will work on during the Lab. They collect analyse and interpret data to address or solve their posed topic. The lab includes an ethical reflection (0.5 EC) on the specific challenge and approach. Fieldwork (3EC) Group size: max 4 During the fieldwork, students will work in teams to define an objective related to the understanding of a phenomenon/process in weather and climate relevant to a societal challenge. Students start with a project planning phase (0.5EC) where they design a measurement experiment. The plan is reviewed and approved by the instructors. In a second phase, the students implement their measurement plan, collect data, process them and analyse the involved uncertainties (1.5EC). Thereafter they will interpret and synthesize their results in a short report and present their findings together with recommendations for future work. (1EC).	
Study Goals		
Education Method	Week 4.1: The purpose/possibilities/scope and boundary conditions of the lab projects will be explained, and the theory components and challenges will be introduced. Students reach out to the teaching staff to inquire about the different lab projects. At the end of this week, students should have a first draft formulation of the Lab objectives and a list of theoretical units they	

	want to follow. Weeks 4.2 - 4.7: Theory components, in parallel students work on their lab, with bi-weekly review sessions. Compulsory theory units start in week 4.1. All other units start in week 4.2. Weeks 4.8 - 4.10: lab and fieldwork
Assessment	This module has two summative assessments: 1) Labs + theory (80%) Group report + ethical reflection (pass/fail) + oral examination (pass/fail) of theoretical units. Assessed learning objectives: 1 - 9 2) Fieldwork (20%) Group report + oral presentation Assessed learning objectives: 2, 10 - 15
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Atmosphere and Climate Dynamics (discipline core CW) or Module A1, A9, A10 - Time management - Self discipline - Ability to read and understand scientific writing - Ability to write scientific reports - Ability to work in groups, teamwork - Critical thinking - Computer skills (e.g., programming and scripting) - Presentation and public speaking
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Discipline Earth Observation

AESM3002	Earth Observation Technologies	6
Responsible Instructor	Dr.ir. D.C. Slobbe	
Module Manager	Dr.ir. D.C. Slobbe	
Instructor	Prof.dr.ir. R.F. Hanssen	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Contact Hours / Week	0/0/8/0	
x/x/x/x		
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	The goal of this module is to familiarize students with the process of specifying and designing an observational mission/campaign (e.g., a new satellite mission or a ground measurement campaign). This process includes the interpretation and analysis of user needs and their translation to observational requirements, the high-level design of possible technical solutions, and the evaluation of the expected observational performance with respect to user requirements.	
	The module will adopt an inquiry-based learning strategy, where groups of 4-5 students will work throughout the module on an observational mission/campaign motivated by topical scientific or societal issues. Each group of students will be mentored by a domain expert to guide their exploration of available and emerging EO technologies, potential trade-offs in mission/campaign design, the availability of suitable forward and inverse models, etc.	
	The required theoretical background will be provided through a combination of classroom lectures and digital learning. This allows the students to acquire a broad foundation and deepen their knowledge on topics most relevant to their observational mission/campaign.	
	Students will begin by formulating, consolidating, and prioritizing user requirements and translating those to mission/campaign requirements. Students will draw on their prior/existing understanding of the underlying physical principles to select a suitable observation technique to observe the variable(s) of interest. Students will develop a preliminary design for an observational mission/campaign that reconciles user requirements with the capabilities of the selected techniques.	
	Where appropriate, students will be provided with real or synthetic observation data that they will analyse to characterize the observations obtained from their mission/campaign, perform a quality assessment, estimate the parameters of interest and reflect on the ability of the mission/campaign to meet the user requirements. Mission/campaign proposals will be presented in a written report (proposal) and orally (pitch).	
Study Goals	After completing this module, students will be able to:	
	-Formulate user requirements that can be assessed based on parameter estimation [ILO A,B,E]	
	-Evaluate the principles and limitations of generic classes of observation techniques, observation platforms, and data processing techniques [ILO A,B]	
	-Analyse third-party user requirements and translate these into system requirements of an observational system (mathematical and physical) [ILO D]	
	-Design an observational mission to gather the requested observations [ILO A,B,D,E,F,H]	
	-Analyse results from an observation mission by estimating the parameters of interest, including quality assessment [ILO B]	
	-Reflect on the analysis results in relation to third-party stakeholders [ILO E,F]	
	-Contribute to effective group work and communicate orally and in written form on the project results at an academic level [ILO G,I]	
Education Method	Lectures and group practicals/exercises,	
	Quizzes	
	Peer and expert feedback during project sessions.	
Assessment	Module is summatively assessed with a group project, using 1 rubric (similar as thesis project) at the end of the period based on:	
	Assignment report (report includes reflection on LOs)	
	Presentation	
	Oral discussion (may include questions on LOs not directly covered in the assignment work).	
	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Faculty and program core modules	
Academic Skills	-Research skills	
	-Data analysis	
	-Written and verbal communication	
	-Listening	
	-Reasoning	
	-Critical thinking	
Literature & Study Materials	Lecture slides and selected papers	
Judgement	Judgement will be based on rubric (to be developed)	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE):	
	https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

AESM302A	Geo-Data Analysis and Geodesy	9
Responsible Instructor	Dr.ir. A.A. Verhagen	
Module Manager	Dr.ir. A.A. Verhagen	
Instructor	Prof.dr.ing. R. Klees	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr. P.G. Ditmar	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This module targets students interested in learning how to rigorously use geo-data to estimate and monitor changes in the shape of the Earth's surface and its gravity field. The signals of interests can be related to local human activities, such as gas or ground-water extraction, or, for example, related to climate change, such as ice-mass losses in Greenland or Antarctica. In this data-oriented module, students will acquire the skills and theoretical background required to process Earth observation data in order to retrieve the signals of interest, in particular by using Fourier analysis methods. A key aspect of this module is dealing with errors in the data and assessing the quality of the input data as well the estimated parameters. Students will learn how to characterize different types of noise, and to make the best possible estimate of the parameters of interest in presence of noise. By learning about hypothesis testing, students will be able to select the model that best fits a set of measurements. Students will also learn to link the estimated parameters (for example, surface deformation rates) to the underlying geodynamical processes. For this purpose, students will learn about some key geophysical processes, such as Earth tides, glacial isostatic adjustment, and tectonics processes, including Earthquakes. Students will follow this module in combination with Earth Observation Technologies, the Earth Observation discipline core. The module includes three units: Statistical geo-data analysis. Fourier Analysis for Earth Sciences. Geodesy and Geodynamics	
Study Goals	After completing this module, students will be able to: Assess the quality of EO data and derived products Design and apply hypothesis testing procedures to select the model which best represents physical reality Apply spectral analysis techniques to extract relevant geophysical information from EO data Apply geodetic observation and analysis techniques to quantify and characterize changes in the shape of the Earth and its gravity field Analyse the link between geodetic observables and the underlying geodynamical processes Present findings in a precise and organized way, both numerically as well as graphically	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three assignments on unit level (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Expected prior Knowledge	Combination with AESM3002	
Academic Skills	Data Analysis, Visualization	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	Three assignments on unit level (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory). For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	n.a.	
Collegerama	No	

AESM303A	Geo-Data and Geo-Informatics	9
Responsible Instructor	Dr. R.C. Lindenbergh	
Module Manager	Dr. R.C. Lindenbergh	
Instructor	Prof.dr.ing. R. Klees	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This is the module for students interested in exploring, mining and communicating the wealth of relevant information in state-of-the-art geospatial data. Different ways to visualize and process geospatial data, in different formats and projections, on geographic information systems will be explored. It will be discussed how to assess the quality of input Earth observation data, and how this quality propagates through a processing chain towards a quality description of a final product. Methodology will be analysed to assess the spatial-temporal contents of data in terms of repetitive patterns and the scales at which information is present in both the spatial and temporal domain. Finally, it will be discussed how such different information can be extracted from data, and how the significance of the extracted information can be accessed and communicated to different stakeholders in effective and attractive ways.	
Study Goals	After completing this module, students will be able to: Assess the quality of Earth observation data and derived products [ILO B] Design and apply hypothesis testing procedures to select the model which best represents physical reality [ILO A,B] Apply spectral analysis techniques to extract relevant geophysical information from Earth observation data [ILO B] Model and estimate the spatial and temporal variability of Earth observation data and relate it to the underlying geophysical processes [ILO B,C] Select and apply appropriate geo-informatics and machine learning tools to extract, process and communicate information from Earth Observation data [ILO B,D] Present findings in a precise and organized way, both numerically as well as graphically [ILO G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three unit level assignments (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	Combination with DC-EO. Written and verbal communication. Reasoning. Attention to detail. Data analysis.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Simple, non-graphical calculator	
Collegerama	No	

AESM307A	Earth Deformation Processes Across Scales	9
Responsible Instructor	Dr. R.E.M. Riva	
Module Manager	Dr. R.E.M. Riva	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	A. Barnhoorn	
Instructor	Dr. G. Bertotti	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module provides the knowledge and skills to understand, predict and characterise Earth deformation processes from continental (e.g., glacial isostatic adjustments and plate tectonics) towards reservoir scales (e.g., folding, faulting and compaction). Geodetic and geophysical observation techniques will be used to quantify these deformation processes, by extracting physical parameters and assessing their uncertainties. In addition, students will learn to relate the observed movements to subsurface engineering (e.g., resource extraction, storage, tunnelling) or natural processes (e.g., plate tectonics, earthquakes). The module contains three components, 1) Statistical geo-data analysis, 2) Geodesy and Geodynamics, and 3) Geomechanics and Structural Geology.	
Study Goals	After completing this module, students will be able to: Design and apply hypothesis testing procedures to select the model which best represents physical reality [ILO B] Apply geodetic observation and analysis techniques needed to quantify, characterize and explain changes in the shape of the Earth and its gravity field, and changes and expressions of crustal structures [ILO A,B,C] Evaluate the mechanical and deformation response and expressions of rocks to varying stresses within the shallow part of the Earth's crust [ILO B] Analyse the link between geodetic observables and the underlying geodynamical and geomechanical processes from reservoir to global scales, including the effects of subsurface engineering activities [ILO B, D] Present findings in a precise and organized way, both numerically as well as graphically [ILO G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three reports on unit level based on assignments (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-EO or DC-GE.	
Academic Skills	Critical thinking. Use of models and observations to solve practical problems. Awareness of issues related to the use of available observations for the purpose of explaining natural and anthropogenic processes within the solid earth.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	Final grade consists of: three assignments on unit level (25% each) and one module level synthesis assignment (25%). Each component needs to be completed with a minimum grade of 5.0 (out of 10.0). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable.	
Collegerama	No	

AESM309A	Climate Change and Dynamic Landforms	9
Responsible Instructor	Dr. M. Vizcaino	
Module Manager	Dr. M. Vizcaino	
Instructor	Dr.ir. H. van der Marel	
Instructor	Dr. J.E.A. Storms	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. R.C. Lindenberg	
Instructor	Dr. H.A. Abels	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module is aimed at students with an interest in the physics of climate change, its effects on the natural environment and how those can be analysed through geospatial data. In the climate modelling unit, student will learn about the numerical, computational and modelling concepts that underlie general circulation models and coupled climate models, which are the primary tools to explain and/or predict the dynamics of past and future climate. The natural environment is here represented by rivers and deltas, which provide an instructive and societal-relevant study case to learn about the interaction between climate change and the surface of the solid Earth. Finally, the Earth observation unit has a strong hands-on identity, which complements the geological characterization of landforms by teaching how to deliver remote sensing products that can be used in geophysical studies. This module will fit into a study-route that addresses how the natural environment is affected by past and future climate change at decadal to secular scales, by using numerical models of the driving physical processes as well as geological and instrumental observations.	
Study Goals	After completing this module, students will be able to: Analyse the interaction between climate change and the solid Earth's surface, with particular attention to the timescales involved. [ILO A,C] Explain the origin of present-day morphodynamic features by integrating geological and geospatial data. [ILO A,B] Analyse climate simulations for the purpose of understanding the driving physical processes. [ILO C] Analyse source-to-sink sedimentary systems and records in relation to tectonics and climate amongst other parameters. [ILO A] Extract geophysical parameters from geospatial data. [ILO B] Present findings in a precise and organized way, both numerically as well as graphically [ILO A,G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Unit 1-3 assignment on unit level (25% each, 1-dec, 5,8, compulsory) and one module level synthesis assignments (25%, 1-dec, 5,8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM402B	Earth Observation B-module	15
Responsible Instructor	Prof.dr.ing. R. Klees	
Module Manager	Prof.dr.ing. R. Klees	
Instructor	Dr.ir. H. van der Marel	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Instructor	Dr. R.C. Lindenbergh	
Instructor	Ir. F.J. van Leijen	
Instructor	Dr.ir. F. Lopez Dekker	
Instructor	Dr. S.L.M. Lhermitte	
Instructor	B.W. van der Wal	
Contact Hours / Week	week 1: 12 PJ	
x/x/x/x	week 3, 5, 6 : 8 PJ	
Education Period	week 7, 8: PJ	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The module consists of four theory units and a lab. Students can choose between two lab categories, i.e., Geohazards and Climate Impacts. The theory units partially depend on this choice. The unit Time Series Analysis is compulsory, at least two units need to be selected from the following units: InSAR data analysis, GNSS Remote Sensing and Positioning, Optical Remote Sensing, Geostatistical Data Analysis (Geohazards lab) / Climate Data Analysis (Climate Impacts lab). The fourth unit can also be chosen from another discipline (see Brightspace).	
Study Goals	The Lab has a central place in the module, triggering the inquiries and as the place where the students apply the theory to a challenge related to monitoring and/or prediction of climate impacts or geohazards. The student (team, max. 3 students) defines a Lab project related to their chosen challenge. They collect, analyse and interpret data to address or solve their proposed topic. The Lab report includes an individual ethical reflection on the specific challenge and approach. During the fieldwork, students will work in teams and receive the terms of reference as defined by a (virtual) client, with the aim to analyse local to regional deformations and/or mass displacements due to various natural and/or human-induced causes. They need to design a fieldwork campaign accordingly; the design will be reviewed during the project planning review. In the second phase, students collect the data in the field and process historical and newly acquired data. Thereafter they will analyse and interpret the data, present their findings and provide recommendations for future work. After completing the module students will be able to: General: 1. Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. 2. Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. Theory/Lab (module specific): 3. Assess the value and limitations of Earth observation data in addressing a societal challenge related to geohazards or climate impacts. 4. Select Earth observation data suitable for the study of a phenomenon of interest. 5. Design, implement and validate a workflow to derive/extract geophysical parameters from Earth observation data. 6. Quantify and assess the input data uncertainties and the quality of the results. 7. Characterize temporal and spatial variations of geophysical parameters using Earth observation data. 8. Analyse, visualize and interpret findings in a clear and convincing manner. Fieldwork: 9. Plan and design a field campaign that is appropriate for the physical process to be measured. 10. Collect data in the field using different measurement techniques. 11. Explain and quantify the error sources associated with the field measurements. 12. Process and analyse the data collected in the field to give meaningful constraints on the physical process. 13. Effectively communicate with peers, assessors and clients. 14. Contribute to a project as a team player and to the overall project management.	
Education Method	In week 4.1 students will be informed about the purpose/possibilities/scope as well as the boundary conditions to be met in formulating the objectives of the lab. Moreover, the theory units and associated activities will be introduced.	
Assessment	Students make a draft proposal for a Lab project in the first week and will receive feedback on this draft. Moreover, students need to select the theory units.	
Expected prior Knowledge	In week 4.3 they will have to submit the final Lab proposal to be approved by the supervisory team. Students will work on the Lab project from week 4.4 onwards. During that period, lab sessions will be scheduled with the supervisor(s) to review and discuss choices, progress, results.	
Academic Skills	Preparation for fieldwork start in week 4.8, actual fieldwork takes place in weeks 4.9 and 4.10.	
Assessment	Formative assessment is based on exercises for the individual theory units, as well as feedback during the lab project by the supervisors.	
Expected prior Knowledge	The theory units and the lab project are summatively assessed based on a written report and oral discussion. The report is to be handed in at the end of week 4.7. Report needs to include an individual ethical reflection and an individual accountability report (i.e., describing what the individual contributions were). At the end of week 4.8 there will be an oral discussion with the supervisors.	
Academic Skills	The fieldwork is formatively assessed at the midterm; students will receive individual feedback from staff as well as peer feedback from their team members. Summative assessment is based on technical performance as well as project skills (rubric available on Brightspace).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	AESM3002 combined with one of the A-modules AESM302A, AESM303A, AESM307A, AESM309A; or AESM308A (which includes AESM3002)	
Academic Skills	Data analysis, interpretation, visualization	

Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.
Judgement	n/a
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	n/a
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Discipline Geo-Energy Engineering
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AESM3003	Geo-Energy Engineering Applications	6
Responsible Instructor	Dr. A. Barnhoorn	
Module Manager	Dr. A. Barnhoorn	
Instructor	Dr. H. Hajibeygi	
Instructor	A. Daniilidis	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module will provide a general overview of the application areas of geothermal energy, petroleum exploration and production, and energy storage in the subsurface. The students will learn in this core module how these geo-energy application areas contribute to the energy demand of the world, in what way they can contribute to the transition towards a carbon-neutral world, what the opportunities, boundary conditions and consequences are of these applications. The students will learn the basic techniques and principles of reservoir characterization and single phase flow. This core module consists of the following components: Basic principles of reservoir characterization and of single phase flow and how these two complement/interact with each other Introduction into energy transition towards a CO2 neutral future energy production, what is the role of the subsurface. Demand, potential and requirements Basic principles governing and applications of Geothermal Energy (geothermal potential, flow and heat in the subsurface), include role of reservoir characterization and flow Basic principles governing and applications of Subsurface Storage (storage of CO2, energy carriers, boundaries of safe operations, cyclic storage), include role of reservoir characterization and flow Basic principles governing and applications of Petroleum Exploration and Productions, include role of reservoir characterization and flow Basic principles governing and applications of the Effects of Subsurface Engineering (Subsidence, seismicity, changes to reservoir) The course will be a mixture of lectures and assignments to determine the role of the three applications for energy production and how they play a role in for filling demand for energy towards the future. The effect of those operations are included. By doing those individual assignments a conceptual plan will be build that provides the students with an integrated approach to the use of the subsurface for the world's future energy demand.	
Study Goals	After completing this module, students will be able to: Explain what the field of Geo-Energy entails, how the subsurface can be utilised and what its role is in the energy transition. [ILO A] Analyse the basic concepts of single-phase flow and reservoir characterization in relation to geothermal energy, subsurface storage, petroleum exploration and production applications and the effects of engineering in this subsurface [ILO A, C] Evaluate how single-phase flow and subsurface characterisation interact at different temporal and spatial scales [ILO A] Develop a conceptual plan that compares and integrates the different applications utilizing the subsurface for the energy demand towards the future. Quantify the components of the conceptual plan using the physic principles that describe subsurface phenomena. Test this problem towards the sensitivity of different components/issues (flow, mechanics, heterogeneity) [ILO C, D, E, F, H, I] Use written and oral communication skills to effectively exchange results and opinions with researchers, engineers, and Applied Earth Sciences stakeholders. [ILO G, H]	
Education Method	Lectures, assignments. Feedback on intermediate assignment results.	
Assessment	Group assignment (100%)	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Academic Skills	Faculty programme core, AES programme core Academic skills: - group work - effective oral communication - scientific writing - critical thinking, development of concepts - integration of data and disciplines - assess impact and uncertainty	
Literature & Study Materials	All material (handouts, readers etc.) and links to online material will be provided during the course via Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Group assignment (100%)	
Permitted Materials during Exam	n.a.	

AESM304A	Flow and Simulation of Subsurface Processes	9
Responsible Instructor	Dr. D.V. Voskov	
Module Manager	Dr. D.V. Voskov	
Instructor	Dr. J.E.A. Storms	
Instructor	Prof.dr. P.L.J. Zitha	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Dr. H. Hajibeygi	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	In this module, students acquire the necessary tools and knowledge to accurately model the building blocks of subsurface reservoirs and to accurately model flow of fluids/energy through these subsurface reservoirs. The 3 units focus on 1) analyse sedimentary and structural models of how the effective rock properties and the different facies, faults and fractures are distributed in subsurface reservoirs 2) describe, analyse and analytically model flow of fluids and energy through those reservoirs (both single-phase as two-phase fluids) and 3) describe, analyse and model flow numerically through these subsurface reservoirs. The students develop these models themselves, analyses the outcomes of these models and reflect on sedimentary and structural heterogeneities, variations, and uncertainties in these models. Particular attention will be paid to the role of scale of effective properties in these three approaches and how these units can be used to understand, solve and predict the key issues within geo-energy engineering applications such as geothermal energy, subsurface storage, hydrocarbon production and consequences of subsurface engineering.	
Study Goals	After completing this module, students will be able to: Evaluate how to build process and stochastic reservoir models and choose appropriate rock properties or facies distribution [ILO A,C] Analyse the physics and develop an analytical model for two-phase flow, thermal processes and poroelasticity through porous media with different assumptions and for different applications [ILO C] Design a stable and consistent numerical method for modelling of flow and transport in porous rocks [ILO C] Implement and create the three model types (reservoir model, analytical two-phase flow, numerical flow model) with geo-energy related software and codes [ILO A,D] Apply the models to geo-energy related test cases and analyse the solutions with respect to the role of uncertainties, sensitivities, relationships and consequences critically for the different test cases [ILO F] Work as a team on subject related problems and report findings and interpretations, including codes and choices made, in a structured and consistent way. [ILO G,I]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three unit level reports (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-GE.	
Academic Skills	Critical and algorithmic thinking Design an algorithm Writing scientific report	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	Unit reports based on assignments (75% of module grade) Synthesis assignment (25% of module grade) For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	NA	
Collegerama	No	

AESM305A	Characterisation of the Subsurface	9
Responsible Instructor	Dr. J.E.A. Storms	
Module Manager	Dr. J.E.A. Storms	
Instructor	Dr. G. Bertotti	
Instructor	Dr. A. Barnhoorn	
Instructor	Dr. H.A. Abels	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module focusses on the structural and sedimentological architecture of the rocks in the Earth that are important for Geo-Energy Engineering applications. The properties of rocks, as well as their variabilities on all scales, are assessed and quantified. Knowledge of the various processes taking place in the Earth and at the Earth's surface help the students assess and predict subsurface architecture and properties. The units consists of an understanding and modelling of how subsurface reservoirs are build up with respect to reservoir properties and facies distributions. How those changes in properties and distributions can be correlated to variations in sedimentary deposition due to past climate fluctuations within different tectonic settings. How those changes can be complemented within the structural framework in the subsurface (tectonics, deformation, compaction, faulting and folding). Particular attention will be placed to heterogeneity, variations and the issue of scale in subsurface characterization. Furthermore, the students will be taught how process-based understanding of sedimentary and tectonic processes is used to accurately populate subsurface reservoir models and how to accurately predict variations in these properties (both in space and in time).	
Study Goals	After completing this module, students will be able to: Evaluate and quantify process controlling spatial and temporal changes in sedimentological and structural properties of sedimentary successions. [ILO A,C] Evaluate and quantify state of stress in sedimentary basins, the way rocks deform and the mechanic factors controlling their response to geo-energy activities. Predict spatial and temporal changes of stress, strain and mechanic properties in the subsurface. [ILO A,C] Interpret the various sedimentary successions and structural features from theory, field, seismic and borehole data. [ILO A,C] Evaluate how to build sedimentary and structural reservoir models and choose appropriate rock properties and property distributions to characterise the model (including sources of uncertainties) [ILO A] Analyse the origin and scales of heterogeneities in sedimentary deposits and structural features and integrate these into representative elementary volumes [ILO A] Appraise the models in geo-energy related test cases and analyse the solution and investigate the role of uncertainties, sensitivities and relationships for the different test cases [ILO D,F] Work as a team on subject-related assignments and report findings and interpretations, including codes and choices made, in a structured and consistent way. [ILO G,I]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three unit level reports (25% each, 1-dec, 5,8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5,8, compulsory).	
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Literature & Study Materials	Combination with DC-GE.	
Judgement	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM307A	Earth Deformation Processes Across Scales	9
Responsible Instructor	Dr. R.E.M. Riva	
Module Manager	Dr. R.E.M. Riva	
Instructor	Dr.ir. H. van der Marel	
Instructor	Prof.dr.ir. P.J.G. Teunissen	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	A. Barnhoorn	
Instructor	Dr. G. Bertotti	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module provides the knowledge and skills to understand, predict and characterise Earth deformation processes from continental (e.g., glacial isostatic adjustments and plate tectonics) towards reservoir scales (e.g., folding, faulting and compaction). Geodetic and geophysical observation techniques will be used to quantify these deformation processes, by extracting physical parameters and assessing their uncertainties. In addition, students will learn to relate the observed movements to subsurface engineering (e.g., resource extraction, storage, tunnelling) or natural processes (e.g., plate tectonics, earthquakes). The module contains three components, 1) Statistical geo-data analysis, 2) Geodesy and Geodynamics, and 3) Geomechanics and Structural Geology.	
Study Goals	After completing this module, students will be able to: Design and apply hypothesis testing procedures to select the model which best represents physical reality [ILO B] Apply geodetic observation and analysis techniques needed to quantify, characterize and explain changes in the shape of the Earth and its gravity field, and changes and expressions of crustal structures [ILO A,B,C] Evaluate the mechanical and deformation response and expressions of rocks to varying stresses within the shallow part of the Earth's crust [ILO B] Analyse the link between geodetic observables and the underlying geodynamical and geomechanical processes from reservoir to global scales, including the effects of subsurface engineering activities [ILO B, D] Present findings in a precise and organized way, both numerically as well as graphically [ILO G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Three reports on unit level based on assignments (25% each, 1-dec, 5.8, compulsory) and one module level synthesis assignment (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-EO or DC-GE.	
Academic Skills	Critical thinking. Use of models and observations to solve practical problems. Awareness of issues related to the use of available observations for the purpose of explaining natural and anthropogenic processes within the solid earth.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	Final grade consists of: three assignments on unit level (25% each) and one module level synthesis assignment (25%). Each component needs to be completed with a minimum grade of 5.0 (out of 10.0). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable.	
Collegerama	No	

AESM309A	Climate Change and Dynamic Landforms	9
Responsible Instructor	Dr. M. Vizcaino	
Module Manager	Dr. M. Vizcaino	
Instructor	Dr.ir. H. van der Marel	
Instructor	Dr. J.E.A. Storms	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. R.C. Lindenberg	
Instructor	Dr. H.A. Abels	
Instructor	Dr. S.L.M. Lhermitte	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module is aimed at students with an interest in the physics of climate change, its effects on the natural environment and how those can be analysed through geospatial data. In the climate modelling unit, student will learn about the numerical, computational and modelling concepts that underlie general circulation models and coupled climate models, which are the primary tools to explain and/or predict the dynamics of past and future climate. The natural environment is here represented by rivers and deltas, which provide an instructive and societal-relevant study case to learn about the interaction between climate change and the surface of the solid Earth. Finally, the Earth observation unit has a strong hands-on identity, which complements the geological characterization of landforms by teaching how to deliver remote sensing products that can be used in geophysical studies. This module will fit into a study-route that addresses how the natural environment is affected by past and future climate change at decadal to secular scales, by using numerical models of the driving physical processes as well as geological and instrumental observations.	
Study Goals	After completing this module, students will be able to: Analyse the interaction between climate change and the solid Earth's surface, with particular attention to the timescales involved. [ILO A,C] Explain the origin of present-day morphodynamic features by integrating geological and geospatial data. [ILO A,B] Analyse climate simulations for the purpose of understanding the driving physical processes. [ILO C] Analyse source-to-sink sedimentary systems and records in relation to tectonics and climate amongst other parameters. [ILO A] Extract geophysical parameters from geospatial data. [ILO B] Present findings in a precise and organized way, both numerically as well as graphically [ILO A,G]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	Unit 1-3 assignment on unit level (25% each, 1-dec, 5,8, compulsory) and one module level synthesis assignments (25%, 1-dec, 5,8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM403B	Geo-Energy B-module	15
Responsible Instructor	Prof.dr. P.L.J. Zitha	
Module Manager	Prof.dr. P.L.J. Zitha	
Instructor	Dr.ir. G.G. Drijkoningen	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr. J.E.A. Storms	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. P.G. Ditmar	
Instructor	Prof.dr. L.G. Evers	
Instructor	A. Barnhoorn	
Instructor	Dr. G. Bertotti	
Instructor	Dr. H.A. Abels	
Instructor	Dr. A.W. Martinius	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. A.M.H. Pluymakers	
Instructor	Dr. A.G. Muntendam-Bos	
Contact Hours / Week x/x/x/x	week 1: 12 PJ week 3, 5, 6 : 8 PJ week 7, 8: PJ	
Education Period	4	
Start Education	4	
Course Language	English	
Course Contents	The lab and the theory part of the B-module are connected and inquiry-based. Theory (8EC) The module includes the following theory components: Compulsory component (2 EC): Induced Seismicity Subset 1 components (at least 4 EC): Geophysical prospecting (2EC), Geological interpretation of seismic data (2EC), Production science and engineering(2EC). Subset 2 components (at most 2 EC): Geostatistical data analysis, Exploration tools and methods, Advanced resources modeling, Time series analysis, (In)SAR, Optical remote sensing, Climate data analysis, Sea level change extremes A short description of each theory component follows: Geophysical prospecting: · Formulate the conceptual relationships between physical properties of the subsurface and potential-, diffusive-, and wave-field measurements. · Explain how these properties can be retrieved through processing or inversion of the collected data. · Choose and apply the proper method(s) to solve a specific problem related to Energy Transition or Geohazard applications. Geological interpretation of seismic data: · Explain what information is present in geophysical data (images and attributes) and to what extent the data can be reliably used for their geological interpretation and apply this to the geophysical data sets. · Provide a geological interpretation of the structure, the sedimentary environment, and the properties of the rocks imaged in a geophysical data set related to Energy Transition and Geohazard applications. Production science and engineering: · Explain the differences between production technology in hydrocarbon, geothermal and subsurface-storage industry, and evaluate risks and safety issues of its well drilling and operations. · Evaluate production processes and facilities in terms of production scenarios and link them to the effects that engineering applications in the subsurface. Induced seismicity: · Describe and explain the mechanisms of, and the effects that engineering applications in the subsurface have on the natural and build environment and society. The focus will be on induced seismicity and subsidence, and cap rock integrity will be briefly discussed. · Integrate knowledge of ethical issues, international standards and risk management systems into Energy Transition and Geohazards applications. Lab (4EC) Labs are conceived as mini-projects which will be done by groups of 2-3 students. Labs play central role in the module. They trigger the questions on the energy transition challenge or the related geohazards that will be addressed by applying the theory components. The student team can define the topic of their Lab within a predefined energy transition challenge or geohazard theme. Examples of themes include: · Gas depletion from the field · Geological hydrogen or carbon dioxide storage · Extraction of geothermal energy The students collect, analyze and interpret data to address or solve their proposed topic. The Lab includes an ethical reflection (0.5 EC) on the specific challenge. Fieldwork (3EC) Group size: 2-3	

Study Goals	Part 1 (0.5EC): preparation Prepare for the assignment in a team. Discussion of the main scientific and technical challenges related to energy transition and/or geohazards. Read literature on the geological objects investigated. Define the data needed and the strategy to collect them, preparing the necessary (digital) platforms. Part 2 (1.5EC): the field trip In the various outcrops visited, student groups will collect the data necessary to address the questions defined in Part 1. Preliminary data processing will be performed during the late afternoon/evening together with the definition of activities needed for the coming day. Part 3 (1EC): reporting The 3rd part is dedicated to the final processing of the acquired data and the preparation of a schematic report on the first-order questions posed in part 2. General learning objectives for theory, labs and fieldwork.
	After completing the module students will be able to: General: Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. [ILO E,F] Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. [ILO F,G,H] Energy Transition and Geohazards Lab + Theory Define and solve a research topic related to the Energy Transition or Geohazards challenge. [ILO E] Analyse Earth system processes through a combination of data, observations and model outputs (geophysical data, subsurface data, petrophysical data, and monitoring data). [ILO A,B] Extract subsurface characteristics and evaluate the options and limitations of data types for present-day and future societal challenges in Energy Transition or Geohazards. [ILO C,D] Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab project in a clearly written and convincing manner. [ILO F,G] Fieldwork: Design and execute a fieldwork campaign appropriate for the Earth system processes and/or applications to be studied. [ILO B,C,H,J] Describe, identify and measure sedimentary heterogeneities, faults, fractures, folds, and 2D rock property trends at different scales (10-1 to 103m) and qualitatively predict their architecture and their potential impact on dynamic behaviour. [ILO A,B,C] Link field observations of rock property trends at different sub-seismic scales to geophysical data and include the impact of scale to model uncertainty. [ILO B,C] Extract subsurface characteristics and evaluate the options and limitations of data types for present-day and future societal challenges in Energy Transition and Geohazards. [ILO C,D] Present analyses, interpretations and conclusions of the Fieldlab project in a clearly written and convincing manner. [ILO G] Contribute to a project as a team player and to the overall project management. [ILO H,J]
Education Method	Week 1: guided sessions during which the purpose/possibilities/scope as well as the boundary conditions to be met in formulating the objectives of the lab will be explained, and also the theory components and challenges will be introduced. At the end of this week, students have a draft formulation of the questions they want to answer in the lab.
Assessment	Week 2-7: Theory components, in parallel students work on their lab, with bi-weekly review sessions. Weeks 8-10: lab and fieldwork. This module has two summative assessments: 1) Labs + theory (80%) Individual report. Includes a reflection on the activities related to the individual theory components, as well as an ethical reflection. Assessed learning objectives: 1- 6 2) Fieldwork (20%) Assessed learning objectives: 2, 7-12 3) Formative assignments for the theory components in the form of homework, reporting and/or presentations. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Expected prior Knowledge Academic Skills Geo-Energy Engineering Applications (DC-GE) critical thinking, oral presentation skills, written reporting skills

Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	NA
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Discipline Geo-Resources

AESM3004	Economic and Structural Geology	6
Responsible Instructor	Dr. M.W.N. Buxton	
Responsible Instructor	T. Schmiedel	
Module Manager	Dr. M.W.N. Buxton	
Instructor	Dr. G. Bertotti	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This module will equip students with the theory and knowledge to understand the nature, origin and factors controlling primary solid/mineral raw materials in order to understand the physical and chemical characteristics that influence and control the behaviour in terms of recovery of value and utilisation. It consists of two units: 1. Economic Geology in which the characteristics of different ore body styles of mineralisation, the origin, controls, formation mechanisms and abundance of primary and secondary raw materials are explained. 2. Geomechanics and structural geology where the role of structural geology and its significance for the formation of ore bodies are presented.	
Study Goals	After completing this module, students will be able to: Classify the variety of mineral resource commodities of primary/secondary origin and discuss the relevance to society. [ILO A] Illustrate the spatial and temporal distribution of mineral resources in the Earth crust by means of specific examples (Precious metals, base metals, ferrous metals, non-metallics, secondaries). [ILO A] Compare evolutionary concepts about the origin of mineral deposits by means of specific examples (Precious metals, base metals, ferrous metals, non-metallics, secondaries). [ILO A,C] Classify different mineral deposit styles based on geological features. [ILO A,C] Relate different processes that give rise to different ore deposit styles to their genetic classification based on the observed geological features. [ILO A,C] Define rock stress and strength and how rocks respond to different styles of deformation. [ILO A] Reconstruct the structural evolution of multiscale rock bodies on the basis of documents such as stress-strain diagrams, seismic sections and outcrop data. [ILO A,C] Appraise the significant processes that lead to formation of an ore deposit during structural deformation. [ILO A,C] Work effectively in a team to define, plan and execute a project assignment and to report the outcomes by means of oral presentation and written report. [ILO G,H,I]	
Assessment	Integrative report on assignment at module-level (50% of module grade), written exam unit 1 (25%, 1-dec, 5.8, compulsory), report on assignment unit 2 (25%, 1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5.8 per Exam/Assessment.	
Expected prior Knowledge	Faculty programme core, AES programme core.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM306A	Extraction Processes and Consequences of Raw Materials	9
Responsible Instructor	Dr. M.W.N. Buxton	
Module Manager	Dr. M.W.N. Buxton	
Instructor	Dr.ir. E.C. Slob	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. G. Giardina	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Ir. M. Keersemaeker	
Instructor	Dr. J. Gebert	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Course Language	English	
Course Contents	This module contains three interrelated units namely Extraction Methodologies, Residual Materials from Post Extraction Processing and the Impact of Primary and Secondary Mineral Raw Materials on the environment.	
Study Goals	After completing this module, students will be able to: Explain the environmental and societal implications associated with the mining related processes to enable the development of innovative management strategies for geo-resources. [ILO C,D] Explain the different extraction methodologies and consequences and implications associated with them. [ILO D] Evaluate the nature of the residual materials from post extraction processing specifically the origin, handling, stacking and storage of mining residues. [ILO B,D] Explain the physical, chemical and fluid Interactions and dynamics for mining residues. [ILO C] Analyse and interpret appropriate geophysical and InSAR monitoring data to detect anomalies and consequences over different time scales of mining residues. [ILO B,C] Translate environmental challenges and economic opportunities in solid waste management into scientific research questions and/or engineering opportunities. [ILO B,C,D,F] Design an integrated plan for the extraction, waste storage and handling, targeting minimal waste and mitigation type strategies for risk/hazards associated with mine waste. [ILO D,E,F] Consider the UN, ICMM and EU goals and regulations when developing engineering solutions for waste management options. [ILO C,F] Work effectively in a team to define, plan and execute a project assignment and to report the outcomes structured and consistent by means of oral presentations and written report. [ILO G,H,I]	
Education Method	See Brightspace for more detailed information on the education method of the module and units.	
Assessment	One integrated summative assignment (design, case study) for all three units (1-dec, 5.8, compulsory). A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Combination with DC-GR.	
Literature & Study Materials	See Brightspace for more detailed information on literature and study materials for the module and units.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESM404B	Geo-resources B-module	15
Responsible Instructor	Dr. M.W.N. Buxton	
Module Manager	Dr. M.W.N. Buxton	
Instructor	Dr.ir. G.G. Drijkoningen	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Dr. M.A. Schleiss	
Contact Hours / Week x/x/x/x	week 1: 12 PJ week 3, 5, 6 : 8 PJ week 7, 8: PJ	
Education Period	4	
Start Education	4	
Course Language	English	
Course Contents	The lab and the theory part of the B-module are connected and inquiry-based. Theory (8EC) [add text] Lab (4EC) Individual or groups of max. 3 students The lab has a central place in the module, triggering the inquiries and as the place where the students apply the theory to a challenge related to characterisation and modelling of mineral resources in order to maximise resource efficiency and/or resource a green future. The student (team) defines a topic related to the challenge within a predefined theme that they will work on during the Lab. They collect, analyse, model and interpret data to address or solve their posed topic. The lab includes an ethical reflection (0.5 EC) on the specific challenge and approach. Fieldwork (3EC) Group size: 3-5 Phase 1: project planning (0.5EC) Teams receive the terms of reference as defined by a (virtual) client, with the aim to analyse specific topic related to resource efficiency and/or resourcing a green future. They need to design a fieldwork campaign accordingly; the design will be reviewed during the project planning review. Phase 2: data collection & processing (1.5EC) Data collection in field and processing of historical and newly acquired data. Phase 3: analysis & interpretation (1EC) Data analysis and interpretation, answering the questions from the terms of reference, reporting in form of a final group presentation and a written report.	
Study Goals		
	After completing the module students will be able to:	
	General:	
	Identify open issues in the processing and/or interpretation of Earth system data records based on the outcomes of the lab project and design a development roadmap to address them. [ILO E,F]	
	Analyse, visualize and interpret findings in a clear and convincing manner. [ILO B,F]	
	Present analyses, interpretations and conclusions, as well as ethical implications, of the Lab and Fieldwork projects in a clear and convincing manner, both orally and written. [ILO F,G,H]	
	Theory/Lab (module specific):	
	Assess the value and limitations of mineral exploration data in addressing a societal challenge related to resource efficiency and responsible sourcing of mineral resources for a green future. [ILO A,B,E]	
	Select appropriate exploration tools and methods and design a workflow to acquire the necessary data for mineral resource characterisation and modelling in order to select a suitable extraction method. [ILO A,B]	
	Analyse and manipulate data to assess the degree of spatial continuity and use deterministic and/or geostatistical interpolation methods to estimate unknown variables of interest. [ILO B]	
	Model the data using advanced geostatistical techniques to define a quantified mineral resource incorporating an estimation of the associated degree of confidence and uncertainty in compliance with international reporting standards (e.g. JORC, PERC). [ILO B,C]	
	Fieldwork:	
	plan a field campaign that is appropriate for a selected deposit style and environmental impact to be measured. [ILO A,D,H,J]	
	collect data in the field using different measurement tools and techniques. [ILO A,J]	
	explain and quantify the error sources and uncertainties associated with the field measurements. [ILO A,B]	
	identify and justify the choice of the selection of extraction methods. [ILO A,B]	
	process and analyse the data collected in the field to assess possible environmental consequences. [ILO B,C]	

	effectively communicate with peers, assessors and clients.[ILO G,H,I]
	contribute to a project as a team player and to the overall project management. [ILO H,J]
Education Method	Week 1: guided sessions during which the purpose/possibilities/scope as well as the boundary conditions to be met in formulating the objectives of the lab will be explained, and also the theory components and challenges will be introduced. At the end of this week, students have a draft formulation of the questions they want to answer in the lab. Week 2 7: Theory components, in parallel students work on their lab, with bi-weekly review sessions. Weeks 8 10: lab and fieldwork.
Assessment	This module has two summative assessments: 1) Labs + theory (80%) Individual report. Includes a reflection on the activities related to the individual theory components, as well as an ethical reflection. Assessed learning objectives: 1-7 2) Fieldwork (20%) Project including a group presentation and a written report. Assessed learning objectives: 3, 8-14 A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	DC-GR
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Master Year 2

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Electives

AESM5110C	Aerosol and Cloud Microphysics	5
Responsible Instructor	Dr. F. Glassmeier	
Instructor	I.M. Steinke	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	.	
Study Goals	After completing this module, you will be able to estimate direct effects of	
	uncertainties related to aerosol- and cloud microphysical parameterization for weather prediction and climate projection inadvertent anthropogenic aerosol emissions and their reductions on weather and climate targeted aerosol injections as part of weather modification and climate-engineering proposals on weather and climate. precipitation on particulate air pollution and to evaluate research literature on detailed quantifications of these effects. You will specifically be able to: Mathematically model important source, sink and transformation processes of liquid and frozen cloud water and atmospheric aerosol particles. Design, perform and interpret numerical experiments and predictions with an aerosol-cloud parcel model. Summarize and evaluate the limitations, challenges, and uncertainties of predicting aerosol and cloud microphysics and aerosol cloud interactions.	
Education Method	lectures (semi-weekly) tutorial class (weekly) homework assignments (pen and paper calculations, numerical simulations, weekly during first part of the course)	
Assessment	final project (paper discussion + analytical calculations or numerical simulations to evaluate its limitations and uncertainties) Assessment (Formative): in-class polling homework correction/discussion pre-report presentation of final project Assessment (summative) Written report that documents the final project (100%)	
Expected prior Knowledge	A minimum of 5.8 per exam/assessment is required to pass this course. Essential: introductory physics and mathematics, esp. thermodynamics and simple differential equations; recommended: introductory Earth and/or environmental science; students from Applied Physics, Aerospace Engineering or similar are welcome (please have a look at the suggested textbooks to judge the match with your background)	
Academic Skills	-	
Literature & Study Materials	Lamb & Verlinde: Physics and Chemistry of Clouds (selected sections, available as e-book at TUD library) Lohmann, Lüönd, Mahrt: An introduction to clouds from the microscale to climate (selected sections, available as e-book at TUD library) comprehensive aerosol-cloud parcel model with ipython notebook interface selected research articles	
Judgement	-	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

AESM5120C		Data Assimilation for Geosciences	5
Responsible Instructor	Ir. F.C. Vossepoe		
Instructor	Prof.dr.ir. A.W. Heemink		
Instructor	Prof.dr.ir. M. Verlaan		
Instructor	Dr.ir. R.W. Hut		
Contact Hours / Week x/x/x/x	5/0/0/0		
Education Period	1		
Start Education	1		
Exam Period	1 2		
Course Language	English		
Course Contents	The course contains the following topics: Bayesian formulation of data assimilation Variational methods (4DVar) Ensemble smoothers (EnRML, ESMDA) Ensemble Kalman Filter (EnKF) Particle Filter Applications of data assimilation in oceanography, meteorology, hydrology, seismology, reservoir engineering and/or geotechnics; Open and FAIR science, open interfaces		
Study Goals	1. To explain the Bayesian principles of data assimilation 2. To relate selected data assimilation methods to these principles (Strong-constraint 4Dvar/Weak-constraint 4Dvar/EnRML, ESMDA/EnKF/Particle Filter) 3. To evaluate existing data-assimilation methods and select the most appropriate method for a given data-assimilation problem 4. To design a data-assimilation workflow that connects a selected model of a particular earth system (oceanographic, meteorologic, hydrologic, reservoir engineering, seismology, geotechnical) to a data assimilation method, or of another estimation problem (traffic, medical sciences, finance) 5. To apply open interfaces 6. To apply a data-assimilation method to a proposed data-assimilation problem		
Education Method	Lectures (2 hours per week) and practical assignments (3 hours per week), self-study		
Course Relations	Can be combined with the Applied Mathematics course Data Assimilation WI4475 on algorithms for data assimilation. The course also links to the cross-over course Advanced Topics in Probabilistic and Statistics for Engineeris in the second quarter of year 2 of the MSc programmes.		
Assessment	Assessment (Formative) In-class assignments Assessment (summative) Report of an individual assignment (application of data assimilation in a problem selected by the student) (100%) A minimum of 5.8 per exam/assessment is required to pass this course.		
Remarks	Which programmes/tracks can follow this module? AES, EE, CT		
Expected prior Knowledge	Knowledge on statistics and partial differential equations as taught in Modelling, Uncertainty and Data for Engineers (CEGM1000)		
Academic Skills	-		
Literature & Study Materials	Data Assimilation Fundamentals, A Unified Formulation of the State and Parameter Estimation Problem, by G. Evensen, F.C. Vossepoe and P.J. van Leeuwen, open access: https://link.springer.com/book/10.1007/978-3-030-96709-3 and selected publications (online)		
Judgement	-		
Permitted Materials during Exam	-		
Collegerama	No		

AESM5210C	Climate Remote Sensing	5
Responsible Instructor	Dr. P.G. Ditmar	
Instructor	Prof.dr.ing. R. Klees	
Instructor	Dr.ir. B. Wouters	
Instructor	Dr.ir. D.C. Slobbe	
Contact Hours / Week x/x/x/x	6 h / week (weeks 1-6); 2 h / week (weeks 7-10);	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The module is set up to meet the growing public concern about climate change and potential shortage of freshwater resources, which are consumed with an increasing rate. To address these concerns, it is vital to measure, unravel, forecast, and, if possible, control various climate- and water-related processes. In this context, the data delivered by remote sensing techniques play a central role. In this module, students will learn how to process, invert, and combine data, including those of different spatial/temporal resolutions. Moreover, they will learn how a proper data weighting can be implemented in order to optimize results of data combination. Using data provided by satellite altimetry and satellite gravimetry missions as an example, students will be able to produce state-of-the-art estimates of large-scale mass change in various Earth system compartments, such as the melting of glaciers and ice sheets, climate-driven variability in river basin water storage, and depletion of groundwater stocks.	
Study Goals	After completing this module, students will be able to: 1. Design a scheme for level-2 satellite gravimetry and satellite altimetry data processing, taking into account data uncertainties 2. Design stand-alone and joint inversion schemes that take into account stochastic models of data errors, as well as possible inconsistencies between datasets 3. Assimilate data into a simple numerical model and assess the added value of data assimilation 4. Apply stand-alone and joint inversion schemes to quantify and separate processes at different time scales in the context of cryosphere, ocean, or hydrology 5. Summarize and defend the findings of a conducted study of selected Earth system processes	
Education Method	Lectures, in-class assignments, individual weekly homework assignments, individual integrated project at the end (combined with an oral presentation and a written report)	
Course Relations	The students will benefit from basic knowledge of surface spherical harmonics, which are taught, e.g., in course AESM2001 (Physical principles of Earth System Observation)	
Reader		
Assessment	Assessment (formative): Guiding in-class assignments (feedback in real time) Providing feedback to (timely) completed weekly homework assignments Assessment (summative): Portfolio composed of: (100%) Completed homework assignments Oral presentation Report on the integrated project (no exam at the end) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module?	
Expected prior Knowledge	MSc AES, MSc CE, MSc EE Basic knowledge of Python language Parameter estimation in linear models Basics of statistical theory Surface spherical harmonics (recommended)	
Academic Skills	Conceptual thinking, critical thinking, problem solving and creativity, communicating and presentation skills	
Literature & Study Materials	Lecture slides, lecture notes, online material for further reading (e.g. scientific articles)	
Judgement	-	
Permitted Materials during Exam	N/A	
Collegerama	No	

AESM5220C	Microwave Remote Sensing of the Earths Surface	5
Responsible Instructor	Dr. S.C. Steele-Dunne	
Instructor	Dr.ir. B. Wouters	
Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	By the end of this course, students will be able to work with, and perform research based on real or synthetic data from current and imminent microwave missions (e.g. Sentinel-1, ROSE-L, CIMR, Harmony, Metop ASCAT/SG-SCA) to study processes relevant in land, ocean and cryosphere applications, in particular processes related to surface-atmosphere interactions. Microwave dielectric properties of natural materials Microwave remote sensing of soil and vegetation. The relation between surface soil moisture, root zone soil moisture and vegetation water content variations and their role in land-atmosphere exchanges of water, energy and carbon. Modelling the influence of dielectric properties and geometry on emission and scattering from vegetated surfaces. Soil moisture estimation from passive and active microwave remote sensing. Monitoring biomass and vegetation water status using passive and active microwave remote sensing. Land-atmosphere interactions over ice/snow The relation between physical snow/firn/ice properties and land/atmosphere interactions for ice The relation between snow/firn/ice properties and the EM properties in the MW region Radiative transfer models to translate snow/firn/ice properties into remote sensing signals for both passive/active MW RS sensors Retrieval of snow/firn/ice properties from MW RS data Ocean The relation between wind, marine boundary layer conditions, and directional surface wave spectra. Theoretical models relating wave-spectra and the resulting directional roughness to the radar scattering intensities and microwave emissivity. Empirical Geophysical Model Functions (GMF) relating surface winds and/or wind stress to radar scattering and Doppler Retrieval of surface wind and surface wave information using data from radar-scatter meters, Synthetic Aperture Radars, and microwave radiometers.	
Study Goals	After completing this module, students will be able to: 1. Explain and describe the relations between surface properties and processes, and observations that can be obtained using microwave remote sensing. 2. Apply state-of-the-art models and retrieval techniques to simulate microwave observables and retrieve states of interest in land, ocean and cryosphere applications . 3. Compare different forward modelling or retrieval techniques to estimate variables of interest at the surface 4. Select and defend the choice of a product/technique/model to capture a process relevant in land, ocean and cryosphere applications.	
Education Method	Video lectures, in-class lectures & discussion	
Assessment	Assessment (Formative) Weekly assignments Assessment (summative) An assignment portfolio, with a final oral defense (100%) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? AES Students from other faculties or universities with an equivalent background can follow this course with approval of the lecturer.	
Expected prior Knowledge	Program Core PPESO or equivalent (check with lecturer)	
Academic Skills	-	

Literature & Study Materials	Lecture notes, online material for further reading (e.g. scientific articles), extracts of text books e.g. Ulaby, Rees, materials provided in class
Judgement	-
Permitted Materials during Exam	-
Collegerama	No

AESM5230C	Coastal Remote Sensing	5
Responsible Instructor	Dr. A. Amiri Simkooei	
Instructor	Dr. R.C. Lindenbergh	
Instructor	Ir. F.J. van Leijen	
Instructor	M.A. de Schipper	
Instructor	A.P. Luijendijk	
Instructor	J.A. Álvarez Antolínez	
Contact Hours / Week x/x/x/x	x/4/xx 4 per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The Coastal Remote Sensing module will provide students with theoretical background on both remote sensing and coastal processes. In this module an overview of coastal processes will be given that are relevant in an applied context, from the coastal shelf to the dunes. It will be discussed at which spatio-temporal scales coastal processes occur, and how these scales relate to remote sensing techniques, from in situ onshore and offshore techniques to satellite systems. An overview will be given of remote sensing techniques for measuring bathymetry and topography, including echo-sounding, laser scanning and photogrammetry, but also the role of spectral techniques will be discussed, to e.g., assess sediment load, sea water temperature and dune vegetation properties. Finally, processing techniques to extract information from data, including data fusion techniques, time series analysis, uncertainty assessment and relevant machine learning methodology will be covered.	
Study Goals	On completion of this module, the student will be able to: Explain which remote sensing techniques are suitable to assess coastal processes in terms of the spatio-temporal scales at which they occur. Develop a workflow to answer a research question on coastal processes using a suitable combination of remote sensing techniques. Extract information from coastal remote sensing data and evaluate the results to answer a research question Associate an error budget to different kind of coastal remote sensing data products and apply this error budget to assess the uncertainty of work-flow outcomes. Defend the results of an implemented workflow.	
Education Method	Lectures. Lectures will be closely supported by dedicated lecture notes, consisting of a mix of existing and newly developed material. In addition, relevant scientific literature will be shared via the lecture notes and Brightspace. During the lectures, students will be engaged by questions (partially hardcoded in the slides, which means that questions are formulated as questions on the slides) and exercises. Instructional videos. Will be shared to shortly explain different concepts on remote sensing and coastal processes. This is also a way to make material available that goes beyond a common core.	
Course Relations	This Module is intended to be closely related to Module CIEM4304 Hydraulics Fieldwork as offered in Q5 within the Track Hydraulic Engineering (Module Manager: Matthieu de Schipper)	
Assessment	Assessment (Formative) Feedback will be organized around informal quizzes, during each lecture Students will be asked each lecture to write down 2 topics they liked and 2 topics that were not sufficiently clear. Assessment (summative) Assignment (100%). Assignment based. In assignments, students will obtain hands-on experience with coastal remote sensing data related to specific coastal processes. Assignments will be organised in groups of ~3 students	
Remarks	A minimum of 5.8 per exam/assessment is required to pass this course. Which programmes/tracks can follow this module? Notably: Discipline Earth Observation within the programme Applied Earth Sciences Track Hydraulic Engineering within the programme Civil Engineering	
Expected prior Knowledge	Modelling, Uncertainty and Data for Engineers (CEGM1000) or equivalent Python programming Below courses/modules/units are not expected prior knowledge, but are related: MSc CE, Track Hydraulic Engineering (HE):	

	A-Module, Hydraulic Engineering Unit: Coastal and Estuarine Systems B Module, Coastal Engineering: Unit: Advanced Coastal and Estuarine Systems, Unit: Coastal and Estuarine Modelling, Loads & Protection MSc AES, Discipline Earth Observation Disc. Core: Earth Observation Technologies A-Modules: Unit, Statistical geo-data analysis Unit, Geo-informatics B-Modules: Unit: GNSS Unit: (In)SAR Unit: Optical Remote Sensing Unit: (GR + EO): Geostatistical Data Analysis Unit (Track: Climate & Weather): Sea level change and floods
Academic Skills	Teamwork, independency, problem solving, creativity, flexibility, time management, communication skills, critical thinking, analytical skills
Literature & Study Materials	Lecture notes Books, with a preference for books freely available to students as eBook. Relevant titles include: Tiberius, C. C. J. M., van der Marel, H., Reudink, R. H. C., & van Leijen, F. J. (2021). Surveying and Mapping, https://doi.org/10.5074/T.2021.007 Bosboom + Stive, specific chapters from Coastal Dynamics. https://doi.org/10.5074/T.2021.001 Rees, W. G. (2013). Physical principles of remote sensing. Cambridge university press. Vosselman, G., & Maas, H. G. (2010). Airborne and terrestrial laser scanning. CRC press. Lurton, X. (2002). An introduction to underwater acoustics: principles and applications. Springer Science & Business Media. Short instruction videos
Judgement	-
Permitted Materials during Exam	N/A
Collegerama	No

AESM5240C	Applied Space Geodesy	5
Responsible Instructor	Prof.dr.ir. R.F. Hanssen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Satellite Earth Observation and Space Geodesy is an increasingly fertile field, with exciting technology that leads to a wide range of applications. Pioneers in the field are often young entrepreneurial start-ups that provide solutions that are directly applicable in society. Examples include, e.g., Planet, Maxar, Iceye, and Capella. What makes those initiatives so successful, and how can we develop new opportunities and challenges that have high impact? In this course you will create your own start-up based on Satellite Earth Observation. We discuss, e.g., core space technologies, value proposition design and strategies, data processing, funding, IP, scalability, valuation, and collaborate as a team to develop a new Earth Observation product or service. We evaluate current European policies to stimulate value creation from Earth Observation, and review legal restrictions. We will use guest lectures from space industry leaders and entrepreneurs as well as site visits to learn about the challenges and pitfalls. Our information products will be used to respond to scientific or societal challenges. Examples include information products for the Dutch Urban Search and Rescue (USAR) teams, requiring emergency response information, or energy sustainability for solar farms.	
Study Goals	After completing this module, students will be able to: Evaluate the space technologies available for value creation Develop an idea into an entrepreneurial business model and a new Earth Observation product or service Create a start-up up to a Minimal Viable Product	
Education Method	In-class lectures, video lectures, site visits, and discussion	
Assessment	Assessment (Formative) Weekly assignments Assessment (summative) Final written report and presentation (100%) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? Aerospace Engineering, Computer Science, Geomatics, Civil Engineering, Applied Earth Science	
Expected prior Knowledge	-	
Academic Skills	-	
Literature & Study Materials	Lecture notes, online material for further reading (e.g. scientific and professional articles), extracts of text books, materials provided in class	
Judgement	-	
Permitted Materials during Exam	N/A	
Collegerama	No	

AESM5310C	Geo-Energy Integration Project	10
Responsible Instructor	A. Daniilidis	
Instructor	Dr. R. Ghose	
Instructor	Dr. K.H.A.A. Wolf	
Instructor	Prof.dr. P.L.J. Zitha	
Instructor	Dr. H. Hajibeygi	
Instructor	Prof.dr. D.F. Bruhn	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. P.B.R. Bruna	
Instructor	Dr. A.G. Muntendam-Bos	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	On the basis of subsurface data of a given area, one subsurface application is selected (CO2 storage, petroleum, geothermal, or other subsurface engineering). A development plan will be formulated for the chosen application and the project will be described from appraisal to full field development. The resulting development plan will be presented in written and oral form.	
Study Goals	After completing this course you are able to: Apply acquired knowledge of the first year MSc on a realistic field study Explain the interaction between the specialist disciplines in Geo-Energy Engineering and deal with inaccuracies Outline the implications of subsurface activities Integrate specific knowledge from geophysics, petrophysics, geology and reservoir engineering for setting up a subsurface development plan using seismic, petrophysical and reservoir data Quantify uncertainty in subsurface development Compile contributions from different disciplines as a team Design a subsurface development plan using seismic, petrophysical and reservoir data Present a subsurface development plan to a management panel Justify choices made for a subsurface development plan (to peer review and teaching team)	
Education Method	In-class lectures, video lectures & discussion, Assignment	
Assessment	Assessment (Formative) Progress presentation, in-class discussion Assessment (summative) Group Report on application design (70%) Group Presentation on application design (30%) Assessment is group-based. Groups have typically 2-3 members. A development report will be written by each group, and presented to a panel. A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? Geo-Energy track students	
Expected prior Knowledge	Students are eligible to take this course in case they have passed the following courses of this track: Geo-Energy Engineering Applications (AESM3003)	
Academic Skills	Peer group work, Integration of knowledge, report writing, presenting	
Literature & Study Materials	Lectures during the course, and additional material to be provided	
Judgement	-	
Permitted Materials during Exam	-	
Collegerama	No	

AESM5320C	Outcrop Geology for Subsurface Characterization	5
Responsible Instructor	Dr. G. Bertotti	
Instructor	Dr. J.E.A. Storms	
Instructor	Dr. A. Barnhoorn	
Instructor	Dr. H.A. Abels	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Module I (1 EC): preparation activities.	
	Students will gather essential information on the region (including geological maps and sections, and literature on relevant processes) and place them in a digital platform such as Google Earth. They will prepare knowledge and tools needed to address the tasks tackled during fieldwork and design the optimal strategy.	
	Module II (3.0 EC): Fieldwork	
	During the first 1-2 days, all participant will be exposed to the area of study defining together the main sedimentary and structural issues. In the following 6-7 days students will work on the tasks which have been defined during Module I. These tasks consist of measurements, observations and interpretations on relevant topics and will last 2-3 days each, which means that each group will have 2-3 tasks to perform. Gathered data will be processed as much as possible during the evening in order to guide data acquisition in the following days. The acquired data and their preliminary interpretations will be handed in to the course instructors. During the final 1-2 days, the different groups will present their findings in the field to the other participants including those of different specializations.	
	Module III (1EC): Finalization	
	During module III, students will further process and interpret the data gathered in the field to reach higher-level conclusion which allows them to predict properties of subsurface rock bodies. This will result in a schematic report. An important part of module III is the organization of gathered information in a data base which can be used by students of following years.	
Study Goals	After completing this module, students will be able to:	
	1) Describe and classify in 3D sedimentary rocks and heterogeneities at various scales in the field	
	2) Describe and classify in 3D faults, fractures and folds at various scales in the field	
	3) Quantify uncertainty in geomodels based on field observations	
	4) Choose predicting and upscaling strategies base on field observations	
	5) Present results in a compact and clear manner	
	Link to intended learning outcomes programme:	
	Observe, characterise and explain Earth System processes (ILO A)	
	Develop and apply data processing and analysis techniques to analyse Earth System processes (ILO B).	
	Model and predict Earth system processes and their variability, and assess the influence of natural and anthropogenic factors (ILO C)	
	Develop novel engineering solutions to facilitate the exploitation and/or management of Earths natural resources in a responsible and sustainable way (ILO D).	
	Formulate a research question, perform a literature, and research study, and build on existing technologies from different disciplines needed for an Applied Earth Sciences solution (ILO E)	
	Challenge existing knowledge, show a critical attitude, and produce creative, constructive and novel solutions, and exercise independent judgement and uphold ethical standards (ILO F)	
	Use written and oral communication skills to effectively exchange results and opinions with researchers, engineers, and Applied Earth Sciences stakeholders (ILO G).	
	Set up, plan and monitor a project, dealing with a deadline and requirements set by Applied Earth sciences Stakeholders (ILO H).	
	Work effectively in a team of diverse talents, skills, characters, and cultures to solve an Applied Earth Sciences challenge (ILO I).	
	Design and execute a fieldwork campaign for the application and/or Earth system processes to be studied (ILO J).	
Education Method	Module I will take place in Delft and will be a combination of practical work supervised by staff and self-study.	
	Module II will be in the field	
	Module III will take place in Delft and will be a combination of practical work supervised by staff and self-study.	
Assessment	Assessment (Formative)	
	Feedback during the preparations, during the field days and during evening Q/A sessions	

	Assessment (summative)
	Report (85%); engagement during fieldwork, participation to discussions etc (15%)
	A minimum of 5.8 per exam/assessment is required to pass this course.
Remarks	Which programmes/tracks can follow this module?
	AES MSc program
Expected prior Knowledge	Module AESM305A
Academic Skills	-
Literature & Study Materials	To be shared prior to and during class
Judgement	-
Permitted Materials during Exam	-
Collegerama	No

AESM5410C	Occupational Health and Safety Management	5
Responsible Instructor	Ir. M. Keersemaker	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course includes the following topics: - Safety systems - Safety culture - Statistics in legal, health and safety - History of health and safety - Safety structure: Organisation - Safety processes: Standards, policies and procedures, - Risk assessment and management, - Policy support and reinforcement, - Toolbox topics - Job safety analyses - Safe working procedures - Health topics: Fatigue management, Noise - Incident investigation - Legal framework - Environmental aspects associated with LHS	
Study Goals	After completing this module, students will be able to:	
	1. explain and describe the importance of Health and safety management in the resources industry 2. find and apply legal requirements and applicable international standards 3. apply knowledge of safety procedures and standards in the field 4. prepare a job hazard analysis and risk assessment and develop mitigation and safety procedures 5. set up an incident investigation	
Education Method	In-class lectures, video lectures & discussion	
Assessment	Assessment (Formative)	
	In-class quiz questions and assignments	
	Assessment (summative)	
	Individual assignment (100%)	
	A minimum of 5.8 per exam/assessment is required to pass this course.	
Expected prior Knowledge	n/a	
Academic Skills	-	
Literature & Study Materials	Lecture notes, online material for further reading (e.g. scientific articles), extracts of text books.	
Judgement	-	
Permitted Materials during Exam	-	
Collegerama	No	

AESM5420C	Geo-Resource Research Driven Project	15
Responsible Instructor	Dr. M.W.N. Buxton	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Ir. M. Keersemaeker	
Instructor	F.S. Desta	
Instructor	T. Schmiedel	
Contact Hours / Week x/x/x/x	10/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Study Goals	The main objectives are: To translate a mining/geo/environmental problem into a relevant research question To design a suitable research plan to investigate the research question according to methodological and scientific standards and within the given time-space To perform research under supervision and develop your research skills To apply appropriate techniques if relevant To handle, analyse, interpret and evaluate data in a correct manner To synthesis and conclude upon the results To define effective communication strategies with the supervisor To write a coherent research paper	
Education Method	Self-study Research driven project work during 10 weeks. The student can choose out of subjects supplied by the Resource Engineering Section at TU Delft or choose a research subject within a mining/mineral resources engineering company. A research subject at mining/mineral resources engineering company is to be agreed upon with the responsible instructor 2 months before starting the course.	
Course Relations	Builds upon knowledge in the Geo Resources Master program	
Assessment	A professional written report (85%). To be finished within the time scope of 10 weeks. A presentation to be assessed by a committee of appointed specialists (15%) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? Geo Resources Master	
Expected prior Knowledge	Have successfully completed Geo Resources MSc core courses (Core, A & B modules)	
Academic Skills	Completion of Bachelors and Masters courses	
Literature & Study Materials	tb Self-Study	
Judgement	-	
Permitted Materials during Exam	-	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Cross Overs

CEGM2000	Suspensions, Sludges and Soils	10
Responsible Instructor	Dr. C. Chassagne	
Module Manager	Dr. C. Chassagne	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr.ir. S.G.J. Heijman	
Instructor	O. Kirichek	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	Dr. J. Gebert	
Contact Hours / Week x/x/x/x	2x4 uur = 8 uur in lecture room 2 x 8 uur = 16 uur practicum	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Sediments and sludges are mixtures of mineral clay, organic material and water. Sediment and slurries are dredged for port maintenance purposes. Sludges found in (waste)water treatment plants are stabilized and dewatered for resource recovery and/or incineration. Some of these slurries/sludges are used in engineered constructions like dikes. In this module, the biochemical and physical properties of sediments and slurries will be reviewed. The way to assess their properties will be studied from the aqueous phase to the solid phase. The module consists in lectures, group assignments and practical work (lab/in-situ). Learning Unit 1: From Suspensions to Slurries EC: 4 Weeks: 1-3 The microscopic/mesosopic properties of clays and organic matter will be reviewed from the point of view of colloid science. The flocculation of particles in combination with different environmental conditions will be studied and modelled. The properties of slurries in terms of settling thickening, rheology and consolidation will be measured and modelled, as function of composition and environmental conditions. List of key topics (Elements of Learning Unit): - Nature of clays (mineralogy) and clay properties - Nature of organic matter and sludge (origin, composition, stability) - Interfacial properties of colloidal particles (surface charge, Derjaguin Landau Verwey Overbeek (DLVO) theory, polymers), - Living organic matter interactions with clay and sludge - Flocculation, coagulation - Rheology of slurries - Settling, thickening, dewatering and consolidation Laboratory experiments and measurements to determine the following properties: - particle size distribution - surface charge - settling rate - coagulation/flocculation - rheological properties - conductimetry - capillary suction time - compressibility - biodegradation of sediment organic matter (SOM) - methane potential of sludges Learning Unit 2: From Slurries to Soils EC: 3 Weeks: 4 - 6 This unit addresses the changes that sediment properties undergo during post dredging treatment and ripening on land. Students will understand and be enabled to engineer the coupled processes of physical, biological and chemical ripening which transform sediments from saturated, soft, anaerobic material to an unsaturated, well-structured and aerated soil fit for use as earthen construction material. Students will understand the associated environmental impacts, for example, related to sediment contamination or degradation of their organic components and the benefits related to sparing excavation of natural soil resources for construction purposes. List of key topics (Elements of Learning Unit): - Mechanical/physical, chemical and biological ripening processes - Methods and criteria to assess physical/mechanical, biogeochemical and biological stability of sediments and soils - Carbon cycling in the sediment-to-soil process - Environmental behaviour of conditioned slurries - Beneficial use of conditioned slurries and sediment Laboratory experiments - geochemical processes during ripening - biodegradation of organic matter - redox potentials - basic soil physical/mechanical and geochemical properties Learning Unit 3: In-situ Characterization of Sediments EC: 3 Weeks : 6-8 The student will learn to use the methods and equipment to monitor the important properties of mud in a natural environment. These properties are assessed for mud in the water phase as well as on land. These methods will be connected to the knowledge acquired in Unit 1 and 2. List of key topics (Elements of Learning Unit): - Monitoring of suspended sediment (submersible laser-diffraction, camera, optical back-scattering equipment, Acoustic Doppler current profiler) - Acoustic methods used for navigation in ports and waterways - Determining in-situ density and flow properties of slurries - Electromagnetic and seismic methods for monitoring of soils and fluid waste repositories	
Study Goals	1. Understand the principal processes that determine behaviour of fine-grained sediments (mud) and sludge in the aqueous phase	

	and in the solid phase. 2. Formulate hypotheses regarding the effect of specific environmental conditions on the behaviour of sediments, sludges and soils. 3. Apply analysis and monitoring methods to study sediments, sludges and soils properties in different environmental conditions (in the laboratory and in-situ) 4. Use the corresponding model(s) to extract the relevant parameters from the raw data obtained from the specific characterization method (point 3) and be able to analyse the results. 5. Assess the properties of mud in relation to a specific engineering problem such as the improvement of water clarity, stabilization of dikes, impact of dredging plumes, navigability of fairways in mud-rich areas.
Education Method	For all units: Interactive lectures (videos, Q&A, discussion about recently published or seminal papers) Targeted self-study using scientific articles, reports or book chapters Laboratory (see description in Units) and field work. Field work will be 1 or 2 days to put theory in practice and will be organized with our partners. Field work is complementary to lab work. At module level: Group project: Case study (cross-over all units) Material and equipment: MUDNET laboratory / sanitary eng. laboratory / Field equipment (partnership) / Geo lab
Assessment	Formative assessment (feedback) Oral feed-back to students (contact hours) will be planned during the whole course, as teachers will be present for demonstrating the use of the equipment, the way to analyse results and the way to report these results. (see in particular LO1, LO2, LO5) Contact hours will also be used to clarify fundamental points that are unclear (in plenary lectures, books, exercises, articles, etc. that students will have to study). The course will be given in the spirit of flipped classroom . Students will be asked to read and comment one or two articles every week. These articles will then be discussed during plenary sessions. The experiments realized during the course of a unit will be formalized in the form of a small report. The design of this report will be discussed and assessed with the teacher, so that students will be prepared for the case study, whereby a full report should be delivered (and that students should design themselves, see summative assessment). During lively discussions, the students will be challenged by the teachers and fellow students to explain their choices and comments their results. The students will acquire the scientific critical mind and their out-of-the-box thinking that is expected from good engineers. Summative assessment Exam each student (60%) At this point, oral examinations are foreseen, which are possible due to the expected amount of students attended the CO module. If this number rises, it will be discussed to have a written exam. During this exam, each student will be asked about the theoretical aspects of the course. Group project report (40%) A case study will be performed and each group of students will deliver a common report on the work that is performed. The case studies will bridge the knowledge acquired in the course of all units. The groups will present the work to an audience and will have to reply to questions. Each student within the group will be challenged, and the final mark for each student will be a weighted coefficient between group report, presentation and individual attitude during the case study and individual response to questions. All subgrades must be > 5,8.
Expected prior Knowledge	N/A
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	All units : State-of-the-art articles (pdfs available on BS) Video recordings (to be recorded available on BS) Slides (pptx) corresponding to the given lectures Unit 1 New Lecture Notes have been made (available on BS), partially based on the book Introduction to Colloid Science application to sediment characterization https://books.open.tudelft.nl/home/catalog/book/16 and several leading publications exercises are being prepared (pdfs available on BS) Chassagne, Claire, Zeinab Safar, Zhirui Deng, Qing He, and Andy Manning. "Flocculation in estuaries: Modeling, laboratory and in-situ studies." In Sediment Transport-Recent Advances. IntechOpen, 2021. handouts for the use of equipment will be prepared and made available Unit 2 Blume et al. (2016): Scheffer/Schachtschabel Soil Science. https://link.springer.com/book/10.1007%2F978-3-642-30942-7 Weil & Brady (2016): The nature and properties of soils. Pons, L. J., & Zonneveld, I. S. (1965). Soil ripening and soil classification : initial soil formation of alluvial deposits with a classification of the resulting soils = Pedogenese initiale et classification des sols = Initiale Bodenbildung und Bodenklassifikation = Pedogenesis initial y clasificacion de los suelos. (Publication / International Institute for Land Reclamation and Improvement; No. 13). Veenman. https://edepot.wur.nl/60013 Journal papers on sediment ripening and beneficial use, exemplary journal: Journal of Soils and Sediments, Springer. https://www.springer.com/journal/11368 Lawrence et al. (2008). Biosolids Engineering and Management, Handbook of Environmental Engineering. ISBN 978-1-59745-174-1 Spinoza and Vesilind (2001). Sludge into biosolids. IWA publishing, London, UK. ISBN-10: 1900222086; ISBN-13: 978-1900222082 Published case studies on beneficial use of sediments: https://www.dredging.org/resources/ceda-publications-online/beneficial-use-of-sediments-case-studies Unit 3 https://library.seg.org/doi/book/10.1190/1.9781560801719 Shakeel, Ahmad, Alex Kirichek, and Claire Chassagne. "Rheology of Mud: An Overview for Ports and Waterways Applications." Sediment Transport-Recent Advances (2021). Draganov, D., Ma, X., Buisman, M., Kiers, T., Heller, K., & Kirichek, A. (2021). Non-intrusive characterization and monitoring of fluid mud: laboratory experiments with seismic techniques, distributed acoustic sensing (DAS), and distributed temperature sensing (DTS). In Sediment Transport-Recent Advances. IntechOpen.
Judgement	N/A

Permitted Materials during Exam	N/A
Collegerama	No

CEGM2001	Sustainable Cities	10
Responsible Instructor	Dr. H.M. Jonkers	
Module Manager	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	approximately 12 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Sustainable cities addresses the need to transform current cities not only to climate mitigating- (substantially lower CO2 and environmental footprint), climate resilient-(adaptive to future climate changes) and circular- (minimal requirement of primary resources and minimal damage to environment) but simultaneously to desirable ones, meaning that they support overall wellbeing of its inhabitants. Some and specifically larger Dutch cities are currently characterized by the presence of aged and desolate city blocks, quarters and infrastructures with often co-occurring environmental and social problems which negatively affect wellbeing of inhabitants. Challenges to overcome are clearly manifold and solutions should therefore be integral in order to be both economically and functionally effective. The indicators that cover the four targeted sustainable (wellbeing) built environment performance criteria (climate mitigation, climate resilience, circularity and wellbeing) and addressed in this module are the following seven: 1. Quality (chemical composition) of outdoors air, water and/or soil (including presence of litter), 2. State of maintenance and quality of present buildings and infrastructure (service life-, circularity-, energy- and environmental performance), 3. Risk of heat stress occurrence (specifically in summertime), 4. Risk of flooding (specifically during intense rain showers), 5. Noise occurrence, 6. Recreational- and social interaction opportunities (quantity and quality of present public spaces) and 7. Quality and quantity of present nature (natural capital) comprising biodiversity and beneficial goods and services that nature can provide to city inhabitants. These seven living quality indicators are specifically addressed in this module as these are considered to be representative for liveability and desirability of sustainable cities. In this Cross Over module specific Dutch city quarters will be analysed both for the qualitative and quantitative state of these seven wellbeing indicators, followed by reviewing available and potential technical solutions for improvement of shortcomings related to individual indicators, and finalized by defining and designing integral solutions for often co-occurring problems. Specific focus of this module will be on the potential of nature-based solutions as these can typically provide integral solutions to multiple climate mitigating-, climate resilience- and wellbeing-related problems. E.g. the Dutch PBL (Netherlands Environmental Assessment Agency) states in their every two-years updated report that more public green (vegetation) in cities is required to increase wellbeing of its inhabitants, and that this can be achieved in balance and in concert with other urgent and space-requiring activities. Latter relate to the societal challenges for the need for more houses and residences, renewable energies, and climate resilience of- and circular building performance in cities. The module Sustainable Cities is practice oriented and case based. Each year specific city quarters will be designated for analysis. Although for practical reasons Dutch cities located close to Delft will be investigated, the problem of insufficient climate mitigation, climate resilience, circularity performance, and lack of wellbeing of inhabitants is relevant for metropolitan cities worldwide. Integral solutions found for Dutch cities are therefore also expected to be applicable to cities worldwide.	
Study Goals	After completion of this module the students are able to: CO-SC-LO1. List and analyse socio-environmental problems occurring in the built environment with specific focus on the air-, water- and soil compartments in city and infrastructural environments CO-SC-LO2. Formulate research and design questions related to socio-environmental problems in the built environment and conduct a multi-disciplinary project in a small group to find answers to the defined question, dealing with deadlines and requirements set in advance CO-SC-LO3. Use written and oral communication skills to effectively communicate theory, facts and results with respect to urban-specific health and socio-environmental problems with researchers and stakeholders in a multi-disciplinary context in order to achieve integral design solutions and strategies CO-SC-LO4. Work effectively in international, multicultural, inter-disciplinary teams CO-SC-LO5. Acquire new knowledge and skills to achieve multi-disciplinary and integral design solutions and strategies for multiple climate-, environmental- and wellbeing related problems occurring in the urban environment CO-SC-LO6. Quantify the environmental impact of linear and circular building practices using LCA (Life Cycle Assessment) techniques CO-SC-LO7. Define the socio-environmental-economic benefits of circular- and nature based building practices using SCBA (Societal Cost Benefit Assessment) related techniques CO-SC-LO8. Develop integral nature-based design solutions for multiple challenges in cities related to the need for climate mitigation and resilience, circularity performance of built structures, and wellbeing of inhabitants	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: - Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes. Small scale experimental demonstrations are given during the lectures in which students participate. - Workshop, hands-on, on campus. Students will learn how to build water-, air-, soil- or noise quality sensors and measuring systems. These will be subsequently used in a practical measuring campaign and obtained results will be analysed and reported. - Visits to designated city quarters is part of the assessment theme. Students will make an inventory of quality of living conditions based on the 7 sustainability performance indicators listed above, and report obtained results. - Case study portfolio. Students will prepare reports of tasks (individual and group tasks) and gather these in an individual case study portfolio. This portfolio will ultimately comprise reports from all assignments covering the module themes assess, review and design. Delivery of assignments will be biweekly formatively assessed. - Case study assignments. Students will form multi-disciplinary groups related to their background expertise during the design theme. Each group will work on a specific set of socio-environmental problems occurring in the designated city quarter and develop multidisciplinary integral solutions based on the knowledge and expertise obtained in the preceding assess and review themes. Groups will report findings to other groups in form of oral presentations. These outputs will be formatively assessed. Final integral solutions proposed by groups will be communicated written (in form of report) and orally (presentation). The final group assignment will be formatively assessed and added to the students individual case study portfolio.	
Assessment	Formative assessment (feedback): During the module, composed of the three themes Assess, Review and Design, tasks have to be delivered. Tasks comprise written reports and oral presentations, in part individual and in part in form of group assignments. The reports will be formatively assessed on completeness and quality by involved lecturers and in one specific case by students in form of a peer-review assignment and feedback will be provided to the students. All tasks will be assembled in an individual case study portfolio which will be summatively assessed for completeness and quality in the last week of the module. Summative assessment:	

	1. One Mobius (computer) individual examination in the last week of the module, covering the theory treated in the given lectures in the three themes (counting for 30% of final grade) 2. The individual case study portfolio will be summatively assessed for completeness and quality in the last week of the module (see Formative assessment, case study portfolio tasks) (counting for 70% of final grade) Minimal grade for both subgrades must be 5.8 to pass the module
Special Information	The Sustainable Cities Cross Over module is divided in the three consecutive and connected themes Assess, Review and Design. A short description of the three themes and its contents is as follows: Module theme 1 Assess: Quantitative assessment of city quarter quality covering 7 performance indicators. In the first theme assess the designated city blocks will be visited and quantitatively analysed for current state of quality based on the 7 indicators listed above: Topics: - Quality of built structures in relation to environmental footprint, circularity- and service life performance- Environmental quality in relation to air-, water- and soil quality and presence of litter and occurrence of noise - Build your own air-, water-, soil- or noise quality sensor (practical; including analysis and interpretation of obtained data and data uncertainty assessment) - Wellbeing quality in relation to risk of heat stress, risk of flooding, quantity and quality of public spaces (recreational- and social interaction opportunities) and quantity and quality of present nature - Methods and techniques for quantification of indicator values- Sustainability assessment methods: LCA (Life Cycle Assessment) and SCBA(Societal Cost Benefit Assessment) Module theme 2 Review: Methods for quality improvement of the built environment In the second theme review specific techniques and methods potentially suitable for mitigation or solving of problems related to the 7 sustainability indicators addressed in module unit 1 will be reviewed. Specific focus will be on green solutions, i.e. through application and implementation of vegetation in the built environment as a way to integrally tackle multiple problems simultaneously. Topics: - Methods and techniques for improvement of air-, water-, soil and noise quality- Circular and other sustainable building strategies for improvement of environmental performance of and wellbeing in city quarters - Eco-engineering solutions for improvement of liveability, climate resilience and climate adaptation of cities. Module theme 3 Design: Integral solutions for improved wellbeing in cities In the third theme design, integral solutions that can significantly mitigate problems, as identified in theme assess, through implementation of technologies reviewed in theme review, will be proposed and quantitatively evaluated. Specifically in theme design students from different MSc program and track backgrounds (EE, AES, CE, Architecture) will work in multidisciplinary teams to apply own background and new obtained knowledge in the assess and review themes to formulate and develop effective nature-based integral solutions to the multiple problems occurring in the analysed city quarters. Specific goal of theme review is to convert current non sustainable city quarters to climate resilient and desirable ones through application of integral sustainable solutions. Topics: - Integral building- and ecoengineering design solutions - Multidisciplinary creation and presentation of design solutions - Critical (qualitative and quantitative) review of design solutions including LCA and SCBA analysis
Expected prior Knowledge	N/A
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	Examples of books: Urban Climates (Oke, Mills and Christen 2017); The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); Stad x Klimaat Het gebouw als watermachine by BNA and TUDelft (2020) (in Dutch, but English handouts will be provided to non-Dutch speaking students); TEEB Manual for Cities: Ecosystem Services in Urban Management (2011) (by The Economics of Ecosystems and Biodiversity (TEEB): Geneva; teebweb.org). These books comprise theory and review a number of practical cases concerning urban heat stress and integral ecoengineering solutions for additional socio-environmental problems occurring in the built environment. Additional specific literature in form of scientific papers and book chapters on analytical techniques will be provided via BrightSpace. Experimental material for the sensor workshop will be provided during the practical.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2002	Engineering for Global Development	10
Responsible Instructor	Dr.ir. M.W. Ertsen	
Module Manager	Dr.ir. M.W. Ertsen	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr. E. Abraham	
Co-responsible for assignments	M.D.M. Palmeros Parada	
Contact Hours / Week x/x/x/x	12 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Focusing on engineering in the Global South, this module relates engineering applications (in global engineering) to a societal debate (what is (desirable) development), a theme spanning domains, programs and tracks within the CEG Faculty. Students learn about similarities and differences of techniques and methods from different domains. Building on the idea that engineering solutions useful and needed as they are are always positioned within a societal debate, students discover their own position. Students learn how to engage their own expertise within debates with other experts and other stakeholders on issues of development, political aspects of engineering, options for interventions, contexts, etcetera. Learning Unit 1: Engineering (and) Development 1 (Credits: 5) In a series of workshops, with discussions, guest lectures, design sessions, etcetera, students are introduced to themes and applications in the Global Engineering domain. After group discussions, individual students prepare documentation on their own position(s). Differences and similarities in debates and positions on technological development influencing engineering solutions are explored. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. These sessions are related to this list of key topics: Development theory (societal development, power relations, economic issues, globalization); Role of technology (engineering decisions, innovation ideas, history of technology); Co-creation and design interventions (participatory design, tinkering, values in design); Sustainable development goals (Differences and connections, development cooperation, role of states and UN institutes). Learning Unit 2: Engineering (and) Development 2 (Credits: 5) Teams of 5 to 7 students work on a real-life issue, relevant to at least a number of different stakeholders (design, planning, methods, etcetera), in the Global Engineering domain. The teams mobilize content from unit 1 in their project, in applying concepts and bringing their team work into unit 1. The projects are made available through the Delft Global network and will vary each year. Examples of possible projects include water provision in urban or rural settings, plastic pollution, affordable housing, access to energy. Team members will discuss the project with members of other teams, to facilitate reflecting on team results. List of key topics: Design teams working on a specific challenge; Team working sessions under guidance of a moderator; Exchange meetings between teams (discussions and workshops); Reporting on team results; Presenting team results in different formats. Please note that the exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. The projects are made available through the Delft Global network and will vary each year.	
Study Goals	After completion of this module the students are able: CO-EGD-1. Explain their definition of and position towards societal development CO-EGD-2. Recognize existing approaches on the role of technology in society CO-EGD-3. Recognize how different stakeholders link engineering solutions to (ideas on) societal development CO-EGD-4. Relate their own expertise to other fields of expertise when developing technologies CO-EGD-5. Activate their own expertise to design engineering solutions for given development issues CO-EGD-6. Specify their own role in processes of co-creation in engineering	
Education Method	Teaching methods involved in Unit 1: Interactive setups, different per session (discussions, role plays, etcetera); Exercises (from closed to more open); (Guest) lectures (a few) Teaching methods involved in Unit 2: Team training (discussion, designing, team roles, team task division); Reporting and presentation skills workshops; Exchange workshops	
Assessment	Formative assessment (feedback): Unit 1: Individual reports: reflection of the learning goals overall in relation to the session; Active participation in sessions (demonstrated in minutes and during sessions) Unit 2: Team: report (quality of content, societal relevance, team work process); Individual: presentations during exchange (quality of content, presentation skills); Individual: debates during exchange between teams Summative assessment: - Unit 1: Individual essay on a theme that relates to the sessions and study material (consistency of the argument, references) - Unit 2: Individual feedback essay on the team project (position in the team, reflection, writing skills) - The final grade is the average of the two grades. Each grade required to be at least 5.8.	
Special Information	Week 1 - week 8: 3 sessions of 4 hours each, with associated reading, group discussions and individual reflection. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. The projects are made available through the Delft Global network and will vary each year. Week 9 - week 10: Self study	
Expected prior Knowledge	This course is open to all students from MSc programs.	
Academic Skills	Analysis of societal positions; reading texts; discussion skills; written and oral communication; team work	
Literature & Study Materials	The study material is collected each year, related to actual debates. Students read papers (about 5 to 7) and discuss these.	
Judgement	N/A	
Permitted Materials during Exam	N/A	
Collegerama	No	

CEGM2003	Data Science and Artificial Intelligence for Engineers	10
Responsible Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Responsible Instructor	Dr. H. Wang	
Responsible Instructor	Dr. R. Taormina	
Module Manager	Dr. R. Taormina	
Instructor	F.P. van der Meer	
Instructor	Dr. K. Anupam	
Instructor	Dr. O. Cats	
Instructor	P.K. Krishnakumari	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. M.A. Schleiss	
Instructor	J.O. Colomes Gene	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	5-7 blocks a week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Elective for MSc programmes AES, CE, ENVM	
Course Contents	Module Introduction Data Science (DS) and Artificial intelligence (AI) are poised to revolutionize all aspects of Civil Engineering, Environmental Engineering and Applied Earth Sciences. Our interdisciplinary module in Data Science and Artificial Intelligence for Engineers (DSAIE) will offer 2nd year Masters students the opportunity to learn these powerful tools and apply them from the very beginning. We will focus primarily on Probabilistic Machine Learning and Deep Learning methods, effective data-driven approaches to deal with the uncertainty and the complexity characterizing the challenges of the interconnected natural, living and built environments. During this module, the students will broaden their engineering understanding through collaboration with peers from diverse backgrounds, following instruction from experts across various programs, and applying what they learn on real-world case studies. The course will also provide plenty of opportunities to further develop their Python coding skills through hands-on exercises and workshops in areas such as collaborative coding, high performance computing, building automation and testing, and advanced data analytics. By the end of the module, the students will have acquired a practical and holistic understanding of DS/AI and its applications in their fields of choice, which will be extremely valuable for the students future careers. Module Units The proposed cross-over module is divided into three units: Unit 1. Data Science & Probabilistic Machine Learning (4 EC) Unit 2. Deep Learning (3 EC) Unit 3. Project (3 EC)	
Study Goals	After completion of this module, the students can: LO1. Explain core concepts in data science, machine learning and artificial intelligence, including the fundamentals of probabilistic and deterministic data-driven techniques. LO2. Translate a real-world problem related to the natural, the living and the built environment into a DS/AI application by following a multidisciplinary approach. LO3. Adapt suitable probabilistic and deterministic DS/AI techniques to solve real-world problems related to the natural, the living and the built environment. LO4. Implement DS/AI pipelines in Python using best coding practices for execution robustness, result reproducibility and validation. LO5. Present the results of a DS/AI analysis in oral and written formats, being able to i) motivate their choices, ii) defend their opinions, iii) cater to diverse audiences, and iv) critically evaluate feedback. LO6. Build workable DS/AI solutions for a multidisciplinary set of realistic problems in cooperation with peers of various backgrounds.	
Education Method	Unit 1 & Unit 2 In a typical lesson, the instructor will illustrate theoretical concepts to the class in a plenary session. When suitable, the instructor will alternate theory with exercises (individual or group) and coding walkthroughs/demos in Python. These will be based on selected datasets chosen by the instructor. While most of the lectures will be on campus, lectures for advanced topics will be delivered via recordings. Unit 3 Theory. Instructors can provide limited content related to the applications via recordings. Advanced topics of UNIT2 can be chosen in combination with the specific project selected. Applications. There will be 3 applications on very diverse datasets for students to apply the concepts of Unit 1 and Unit 2 while preparing for the final project. Students will work in groups of 5; instructors will be available with TAs for support in class. Each application will last for approximately 4 hours, as done in the MUDE. Projects. In week 4, project instructors will introduce case studies/datasets that students can exploit for developing their final projects. Contrary to applications,	

the proposed projects may not have clearly defined research questions. Students will form groups of around 5 to work together on their final projects. Students can also come up with their own dataset and case study, provided they find the support of an instructor that can act as a coach/mentor (open inquiry, see Figure 3). Groups will be formed in Week 5. Students will preferably link the self-study advanced topics of Unit 2 (Week 6) to the project they choose.

Coding Workshops.

The coding workshops will be usually delivered as a short lecture followed by hands on exercises (e.g., on notebooks, HPC terminals, ...) to master specific coding knowledge related to AI and Data Science. Instructors and TAs will be available in class for the entire duration of the workshops.

Practical Guide

Unit 1. Data Science & Probabilistic Machine Learning (4 EC) [LOS 1, 3, 4]
Week 1 to Week 3

In this Unit we will first introduce the module and reactivate prerequisite knowledge from the MUDE with respect to Supervised Regression. We will then explore Supervised Classification, the other major family of Supervised Machine Learning methods. We will then study the Bayesian perspective on model selection, before exploring tree-based methods and kernel based methods for regression and classification. After refreshing MUDE knowledge on clustering and PCA, we will explore more advanced topics in unsupervised machine learning and generative machine learning. Students will work on two different application sessions to reinforce their knowledge.

Unit 2. Deep Learning (3 EC) [LOS 1, 3, 4]
Week 4 to Week 6

This unit will mainly deal with Deep Learning methods, representing the state-of-the-art of AI-based applications in many fields, including CEG-related ones. We will focus on key topics such as representation learning, and the use of inductive biases and symmetries to overcome the curse of dimensionality. These will be introduced with respect to convolutional and recurrent neural networks, but we will explore other powerful and more recent approaches such as physics-based deep learning and transformers. Students will work on one application session to reinforce their knowledge.

In Week 6, students will select two advanced topics to study independently (i.e., videos, materials and notebook). These will be preferably linked to the projects they select in Week 4 (see Unit 3).

Unit 3. Project (3 EC) [LOS 2, 3, 4, 5, 6]
Week 4 to Week 10

Projects are an essential part of the module and constitute a dedicated unit. Students will work in groups on a selected CEG-related project to apply and reinforce the skills acquired in Unit 1 and Unit 2 while collaborating on real case studies. All available projects will be showcased by the responsible instructors in Week 4, and students have time until Week 5 to choose their project.

Potential case studies

Here follows a tentative list of potential case studies from which students can select their projects. Students can choose their projects regardless of their background. Students can also come up with their own project from scratch, provided they find a mentor by Week 5.

Environmental Engineering

Rainfall-runoff modelling using Deep Learning methods
Prediction of drinking water demands
AI-based modelling of water quantity and quality processes in urban water networks
Plastic detection using AI-based computer vision
Defect detection in sewer systems with AI-based computer vision

Civil Engineering

Surrogate modelling for flow through porous structures
Accelerate finite elements methods for structural mechanics with AI.
Defect detection of railway infrastructures using deep convolutional neural networks and on-board camera images.
Flood Risk Mapping with Deep Learning methods.
Unsupervised learning for rail health condition classification.
Unravelling the Dynamics of Urban Demographics
Incorporating safety and beauty perceptions into travel journey planners using street view images

Applied Earth Sciences

Rainfall nowcasting using Deep Learning methods.
Machine Learning for applications in Energy Transitions
One-shot learning for generative modeling of geological structures
Exploring the effect of different formulations of Gaussian processes on spatial predictions

Assessment

Formative assessment (feedback)

Theory

Computer-based quizzes with multiple answers for each topic, including coding exercises. These will be made available every week. A mock-up theory exam will be made available in Week 5.

Project

Formative feedback provided in class during applications 1-3 which serve preparatory activities for the project. Feedback from instructors and peers on project progress at the end of Week 7.

Summative assessment on module level

There will be two summative assessments: 1) an ANS written exam for the theoretical part, and 2) a group presentation with interview for project evaluation.

Students understanding of theoretical content/coding skills will also be assessed as part of the project learning line, specifically their ability to apply the theory and reflect on the meaning and limitations of produced results. This justifies the equal value of the two summative assessments (i.e., 50% each).

The project deliverables include the final presentation plus a code repository. The grade will reflect the originality and impact of the final solution, the quality of presentation, and the quality of code repository and documentation. Dedicated assessment tools for group work will be employed to facilitate marking of individual contributions.

All subgrades must be >5.8 to pass the module.

Special Information

Weekly Syllabus

10 ECs = 280 hours. We consider 28 hours per week divided in 7 blocks of 4 hours each. There will be around 50 hours of lectures, mainly focused on the first half of the module. The second half focuses on the project. Sufficient time is reserved for

self-study during the entire module. The students are free to organize their self-study as they want, but instructors and/or teaching assistants will be available in a classroom for explanations after lectures. When the project runs, there is a block reserved each week to interact with project instructors and TAs.

Week 1

Introduction to the course with overview.

Translating real-world problems to build data science and artificial intelligence solutions.

MUDE Recap: supervised regression, deterministic perspective on model selection: bias-variance trade-off, regularization, cross-validation, neural networks.

Coding Workshop #1: PyTorch

Data types: time series & sequences, tabular data, images & raster data, unstructured vs structured data

Probabilistic perspective on model selection: informative/uninformative priors, hyperpriors, Maximum a Posteriori (MAP) estimates, Bayesian model comparison, Empirical Bayes, Variational Bayes

Week 2

Supervised classification: classification with decision regions, decision theory for classification, classification with basis functions, logistic regression, classification with neural networks

Feature engineering, input selection and dealing with unbalanced data.

Coding Workshop #2: collaborative coding for DS/AI projects.

Frequentist and Bayesian kernel-based methods, instance-based learning

Application #1: Supervised Classification

Week 3

MUDE Recap: unsupervised machine learning, k-means and PCA

Inference and learning with discrete and continuous probabilistic latent variables, introduction to Generative Machine Learning, Probabilistic Principal Component Analysis (PPCA)

Application #2: Unsupervised & Generative Machine Learning

Week 4

From Probabilistic Machine Learning to Deep Learning: curse of dimensionality, representation learning, inductive biases & symmetries for data efficiency

Projects showcase

Convolutional Neural Networks: 1D/2D/3D convolutional layers, pooling, composition, feature extraction, translational invariance and equivariance, data augmentation, image regression and classification.

Regularization: early-stopping and dropout layers.

Pre-trained models and transfer learning.

Coding Workshop #3: AI on DelftBLUE

Application #3: Deep Learning

Week 5

Recurrent Neural Networks: sequential inductive bias, simple RNNs, gated recurrent neural networks (LSTMs and GRUs), time series data.

Intro to Physics-based deep learning: basics of neural ODE/PDE solvers, neural operators, physics informed neural networks, physics guided machine learning.

Other type of inductive biases: quick introduction on transformers, graph neural networks, bayesian deep learning.

Presentation of advanced topics that students can select (see below).

Project selection

Initial work on projects

Week 6

Advanced Topics

Project work

Week 7

Project work

Project preliminary presentations + feedback

Week 8

Project work

Week 9

Project work

Written exam

Week 10

Project work

Project submission and presentation

Expected prior Knowledge

CEGM1000 - Modelling, Uncertainty and Data for Engineers

Academic Skills

Analysis of complex (multi-cause) problems related to the built/natural/living environment; information collection and evaluation; multi-disciplinary and integral problem solving; project management; oral communication; teamwork; basic coding skills.

Literature & Study Materials

Unit 1 & Unit 2

Study material: lecturers slides, recorded videos/Collegerama, Jupyter notebooks, GitHub code repositories + documentation, datasets.

Literature:

Bishop, Christopher M. "Pattern Recognition and Machine Learning", 2006.

Murphy, Kevin P. "Probabilistic Machine Learning: An introduction.", 2021.(open source <https://probml.github.io/pml-book/book1.html>)

Murphy, Kevin P. "Probabilistic Machine Learning: Advanced Topics.", 2022. (open source <https://probml.github.io/pml-book/book2.html>)

Goodfellow, Ian, et al. Deep learning, 2016. (open source <https://www.deeplearningbook.org/>)

Molnar, Christoph. "Interpretable machine learning", 2020. (open source <https://christophm.github.io/interpretable-ml-book/>)

Wilke, Claus O. Fundamentals of data visualization: a primer on making informative and compelling figures, 2019. (open source <https://clauswilke.com/dataviz/index.html>)

Selected seminal papers in Data Science, Machine Learning, Deep Learning and Interpretable Machine Learning.

Select webpages and blogs.

Unit 3

Study material: datasets with description, code repositories, recorded videos (optional), all material of Unit 1 and Unit 2.

Literature: each instructor will provide a selected list of scientific papers and other publications (e.g., MSc Theses of previous

students) for the applications/projects they propose.

Coding Workshops & Computational Resources

The coding workshops will be partially based on high-quality, worldwide known, open-source Data Science and Computing workshops provided by The Carpentries (<https://carpentries.org/>), CodeRefinery (<https://coderefinery.org/>), and Aalto Scientific Computing (<http://scicomp.aalto.fi/training/>). The instructors will select, summarize, modify, and expand the material available from these resources to fit the scope of this Module. Students will be able to access the DelftBLUE supercomputer for the entire module (<https://www.tudelft.nl/en/dhpc>).

Judgement

See assessment

Permitted Materials during Exam

N/A

Collegerama

No

CEGM2004	Noise and Vibration: Generation, Propagation and Effect on Humans and Environment	10
Responsible Instructor	A. Tsouvalas	
Module Manager	A. Tsouvalas	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	approximately 4*2HC Theory and 1*2HC project with instructors per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	General description: Noise is sound considered unpleasant, loud or disruptive to hearing, which has become an increasing concern throughout the world. While urban noise (from transportation, construction, etc.) brings a negative influence on human health, underwater noise (from pile driving, seismic surveys, etc.) threatens the leaving space of marine mammals and fish. Similar to noise, vibration is undesirable in many cases because it causes damage to engineering structures and discomfort to people. From the viewpoint of physics, noise is vibrations transmitted through a medium, such as air or water. Therefore, effective control of noise and vibration can be possibly be achieved at the source, path, or receiver. In this module, students learn the fundamentals of noise and vibration, including structural vibrations, sound generation and transmission, perception of noise by inhabitants, conduct in-class/lab tests to obtain and analyse vibro-acoustic data, and use numerical tools to mitigate the adverse effects on the urban and marine environment. Overall structure of the module: The module structure consists of two interrelated themes; Theory and Project. The Theory theme starts first and students are exposed to fundamentals of noise and vibrations, important sources of noise in air and underwater and dynamics of structures. The coupling between the structure and the surrounding media, in which waves are transmitted, is central in the Theory theme. Vibro-acoustic modelling techniques are discussed and software (that the students use for the Project theme) is introduced. Besides modelling, students are also exposed to measurement setups and equipment regularly used to measure and quantify noise and vibration levels. The theme on Theory is followed by the Project theme in which students apply the theoretical concepts to realistic environmental noise and vibration problems. Case studies, in which levels of vibrations and/or noise are unacceptable, are selected by the instructors together with the relevant industry. Students form groups to tackle the environmental challenging case. Results of this investigation are reported by the group and presented at the end of the quarter. In the theme theory the mathematical background of acoustics and vibration is taught with an eye on real-life applications. Emphasis is placed on an in-depth analysis of structure-borne noise emission and its effective cancellation either at the source or at the propagation path. This unit is developed such that it includes illustrative applications of the different noise sources in the environment including the effects on people and animals. This unit is fundamental and interlinked to the theme project in which a practical implementation of the theoretical concepts is accomplished. In the theme Project, students apply the theoretical concepts to realistic environmental noise and vibration problems. The students are divided into groups. Groups are formed by 2-3 students following different MSc programmes in order to enrich the expertise within the groups and to be able to reach scientific depth in the final result of the project.	
Study Goals	After completion of this module, the students are able to: LO1 Formulate the governing equations for problems in which structural vibrations and sound generation and propagation are involved. LO2 Construct analytical formulae and numerical models to simulate structure-borne acoustic emission. LO3 Analyse the environmental noise and translate those measurements to (internationally) accepted noise metrics. LO4 Evaluate the impact of noise and vibrations on the environment and humans using metrics for noise and vibration perception. LO5 Design noise mitigation strategies to tackle urban and underwater noise pollution. LO6 Compose scientific reports on vibration- and noise-related topics using professional terminology.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process. - Lectures. Pre-study material (book chapters and handouts) will be provided to students before the lectures. The course content will be described with real case studies, examples and interactive demonstrations with the active participation of the students. Students will be asked to refresh their knowledge at parts to avoid repetition of material covered earlier in the curriculum. - Hands-on assignments. While the students learn about the theory and mathematical descriptions related to the physics of noise and vibration, they should apply the outcomes to the case studies. The students should complete bi-weekly tasks in small assignments which will help them to execute the project. Delivery of bi-weekly assignments will be optional and formative feedback will be provided by the instructors. - Workshop. The students will learn the basic noise and vibration measurement equipment, use them, and analyse the collected data. The measurements sessions will be performed either in-class, in the lab facilities or, ideally, in the campus during the workshop. - Self-study: During weeks 1-5, CE students read selected topics related to acoustics and EnvEng, AES students topics that relate to structural dynamics. This way students get exposed systematically to theory from another discipline. The Project theme is organized by means of feedback/plenary sessions. The students form multi-disciplinary groups related to their backgrounds. Each group can choose one of the provided urban or underwater noise problems as a project topic under the supervision of the instructors so that all subjects are covered equally. Each group should develop innovative solutions based on the obtained knowledge in the preceding theory theme. Groups will present their process and results to other groups during bi-weekly meetings with the aim to initiate a live discussion and provide a formative assessment of the work-in-progress. Each group will prepare a detailed final report related to their project. These outputs (presentation and final report) will be summatively assessed.	
Computer Use	The software to be used by the students for the execution of the project will consist of: - available licensed software from TUD (e.g. Comsol Multiphysics) and - in-house software prepared by the teaching team.	
Practical Guide	Weeks 1 and 2: Introduction to noise and vibrations - Fundamentals of acoustics: acoustic metrics, sound wave propagation. - Fundamentals of structural vibrations: vibrations of beams and plates, coupling to the acoustic domain and sound radiation. - In-class examples with analytical solutions, homework.	

	Week 3: Noise and vibration sources and impact on environment (distribution of projects and group forming at the end of week 3) - Noise and vibration sources in the urban and marine environment (traffic, rail, underwater and air noise) - Underwater noise and vibration sources, defining the acoustic problems. - Audibility range of species, levels of injury, etc. - WHO reports, MSFD, biological research, noise metrics, international standards. - Levels of vibrations acceptable in buildings. - Impact of traffic, rail and air-induced noise and vibrations on human life - Underwater noise impact on marine life: Noise from shipping, pile driving, seismic surveys, explosions and wind. - In-class examples with analytical solutions, homework. Weeks 4 and 5: Propagation of waves and modelling - Brief information related to ray theory, mode theory, finite elements, and boundary elements (air/water/soil layered waveguides). - Why do we need simulations? How to choose among the different simulation techniques? - Model validation and interpretation of the modelling results. - Generation of noise maps. - In-class numerical examples (homework related to simulations with FE or inhouse software; the software to be used will be determined by the instructors depending on the type of problem analyzed). Week 6: Sound and vibration measurements - Noise measurement equipment: sound level meters, hydrophones. - Lab-scale (or in-class) demonstration of working principles and data analysis. - Analysis of noise and vibration data, converting pressure to relevant metrics for the noise impact (homework related to the analysis of measurement data). - Tire-pavement measurement methods. - Rail-wheel measurement methods. Weeks 7 and 8: Noise mitigation and vibration suppression - Noise mitigation and vibration suppression strategies, smart mitigation strategies (acoustic black holes, metamaterials), examples for the noise mitigation related to pile-driving. - Measurements in lab/field. - In-class discussion of case studies: noise mitigation and vibration suppression. Project - Conducting a project regarding either urban or underwater noise: Each group will conduct a project (distributed in the end of the 3rd week of the module) with the focus being placed in one of the subjects listed below: 1. Structural-borne noise and vibrations in urban environments: traffic noise including modelling of tire-pavement interaction, railway noise including modelling of rail-wheel interaction, noise from civil construction in urban environments. 2. Structural-borne noise and vibrations in offshore environments: seismic surveys offshore, noise and vibrations from installation of foundations offshore, shipping noise, drilling. To accomplish the project, students are required to conduct simulation, analyse the results of noise and vibration data (data sets will be provided by the instructors), design a strategy of noise mitigation and/or vibration suppression, and compose a final report.
Assessment	Formative assessment (feedback): Students will discuss their results and formative feedback will be provided by the instructors and based on plenary sessions during the quarter on a weekly basis. Formative assessment of weekly assignments regarding theory unit. Summative assessment: The module is assessed by means of the project work with a report and an in-class presentation.
Expected prior Knowledge	N/A
Connection to other modules	The CO module brings together engineers of various backgrounds and challenges them with a problem in which both the source of noise and vibrations, the emission of waves and the mitigation of noise and vibrations are tackled in a holistic manner. This requires teams consisting of students of various expertise which is mobilized during the execution of the project. Given the focus of the module, there is some overlap with the following parts of the individual programmes: - Overlap with the AEE track in the EnvEng programme on the fundamentals of acoustics and measurement of noise; - Overlap with the SE track in the CE programme on fundamentals of structural vibrations and measurements; - Overlap with AES programme on elements of wave propagation. Despite the overlap though, the students here will need to go beyond their comfort zone, tackle a multi-disciplinary problem, and work together to find a suitable remedy to reduce the disturbance to inhabitants. They will also deepen their knowledge further by tackling not only the individual elements dealt with earlier in their programmes but also their complex interaction.
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	Theory: Mathematical description and physics related to noise and vibration, analysis of practical problems of structure-borne noise and vibrations. Selected chapters of the following books: [1] Allan. D Piece, Acoustics: An Introduction to Its Physical Principles and Applications, 2019, Springer. [2] Steven L. Garrett, Understanding Acoustics: An Experimentalists View of Sound and Vibration, 2020, Springer [3] M. Möser, G. Müller, Handbook of Engineering Acoustics, 2013, Springer [4] D. Thompson, Railway Noise and Vibration, Mechanism, Modelling and Means of Control, 2010, Elsevier https://www.sciencedirect.com/book/9780080451473/railwaynoise-and-vibration. [5] F. Fahy, P. Gardonio, Sound and Structural Vibration: Radiation, Transmission and Response, 2008, Academic Press. [6] Robert Bernard and Roger L. Wayson, An Introduction to Tire-Pavement Noise of Asphalt Pavements, 2007. [7] TRBs National Cooperative Highway Research Program (NCHRP) Report 630: Measuring Tire-Pavement Noise at the Source examines a Suggested Procedure for Measuring Tire-Pavement Noise at the Source Using the On-board Sound Intensity (OBSI) Method. Additional specific literature (if any) in form of scientific papers, other book chapters, or public reports will be provided in class. Project: Conducting a project regarding either urban or underwater noise Books: same as theme Theory above. Additional materials including the manuals of the measurement equipment and the papers reviewing the existing studies of the specific topics on urban or underwater acoustics.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	Yes

CEGM2005	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE)	10
Responsible Instructor	E. Ragno	
Module Manager	E. Ragno	
Instructor	Ir. F.C. Vossepoe	
Instructor	G.F. Nane	
Instructor	Dr.ir. O. Morales Napoles	
Instructor	Dr.ir. R.C. Lanzafame	
Contact Hours / Week x/x/x/x	approximately 4 hours lectures, 4 hours workshop per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE) is a methodological type module oriented and structured around advanced topics in probability and statistics applied to engineering problems. Rainfall, water flowing into rivers, cars and trucks crossing bridges, ocean waves, and earthquakes are all examples of phenomena of which we wish to quantify the magnitudes, frequencies, uncertainties, and risks to make our daily life safer and more manageable. Such phenomena are characterized by some degree of randomness, meaning that their future outcomes are unknown. Hence, how do we design sewer systems to deal with precipitation in urban areas, estimate the loads that will compromise the structural integrity of a bridge or protect coastlines from flooding? In this module, students will learn methods of probability and statistics to infer characteristics of random and uncertain phenomena around us when observations (data) and physics-based numerical models are available, by using (non)stationary extreme-value analysis, dependence modelling in higher dimensions and data-assimilation techniques. When data pose challenges or are simply missing, students will learn about structured expert judgment methods to quantify uncertainty by using expert opinion. The module will provide students with a theoretical understanding of different probabilistic techniques so that they can discern which method(s) is(are) most suitable for analysing the problem at hand. At the same time, students will learn how to translate their theoretical knowledge into computer codes. As part of this course, students will apply the methods learned in an interdisciplinary group project around specific themes, that will vary on a yearly basis. Example of projects themes are (i) shoreline protection in the face of sea level rise; (ii) subsidence as a results of ground water extraction; (iii) water management assessment under urbanization and climate change; (iv) reliability of aging infrastructure under climate change. Students will work in groups, about 4 members each, and based will come up with specific research questions for their theme that they will answer using methods learned. The total 10 EC will be divided as follows: 6 EC in Unit 1 - Theory and Example applications and 4 EC in Unit2 - Group Project.	
Study Goals	After completion of this module the students will be able to: LO1 Discuss the concepts of non-stationary extreme value analysis, multivariate probability distributions, multivariate uncertainty quantification, data assimilation, and expert judgment. LO2 - Apply the concepts of non-stationary extreme value analysis, multivariate probability distributions, multivariate uncertainty quantification, data assimilation, and expert judgment to problems in civil engineering, applied earth sciences and environmental engineering. LO3 - Analyse and discuss the engineering applications (for example climate change, extreme discharge, induced seismicity, maximum bending moments, traffic loads, infrastructure design, pollution health effects) that accompany the techniques in probability and statistics taught in this module with an emphasis on the added value of uncertainty quantification. LO4 Write computer codes using open-source, e.g., Python, or commercial, e.g., Matlab, programming languages to translate key methods and techniques presented in the module and to carry out analysis related to specific track-related problems. LO5 Make use of existing open-source code and software to carry out analysis using more advanced methods which require higher-level coding skills. LO6 - Design and carry out an interdisciplinary research project of interest in science or engineering that incorporates methods taught in this module within a guided inquiry-based learning framework. LO7 - Work effectively in international, multicultural, inter-disciplinary team	
Education Method	Theory and example applications Lectures: the module will offer a mix of classical style lectures from an instructor, collaborative group sessions and educational videos. Dedicated classroom time will be provided each week for the lectures and collaborative sessions with students and instructors. Hands-on learning via computer practicum and think-pair-share activities will be implemented in class. The students will benefit from the interaction with their peers and from the need to re-interpreting theoretical concepts in consecutive executable actions. At the same time, teacher-center lectures are necessary to show the students the theory and rationale underlying the statistical methods covered in class. An example application of these activities can be related to extreme value analysis. Students will be provided with a time series for a general variable (for example maximum yearly wave height in the North Sea for CE, maximum PM2.5 concentrations in Amsterdam for EE, or expected probability of earthquake occurrence in AES) and they are asked to extrapolate a sub-sample of the observations containing only extremes. First, they will do this individually and then they will share ideas with their partner. After gathering students ideas, the instructor will explain how extremes can be sampled based on current methods in extreme value theory. Students will then apply such methods into executable coding scripts to sample extremes from a time series of their interest. Applications: Students will work on weekly assignments related to practical applications of the theory learned in class. Such assignments will derive from the three programs so that the students can appreciate the interdisciplinary character of statistical methods. The applications proposed will be related to a specific track in such a way that also students that do not come from that track can understand them, as exemplified in the Module Content section. Students are encouraged to follow the think-pair-share approach. In this way, they will first try to solve the problem individually and then compare their work with their peers for feedback. Dedicated classroom time will be made available for peer feedback and collaboration.	

	Group project A guided inquiry-based learning approach will be followed, similar to that implemented in the MUDE module (Q1 and Q2 of the first year MSc programme). This module builds on the project-related skills developed in MUDE by allowing students to identify their own topic of interest to further analyse it using at least two methods covered in class. Students progress will be followed by the instructors starting from week #4, in which an introduction on the remaining topics covered in the module will be given to facilitate the project definition. Guidance documents for students (e.g., effective group work strategies, formulation of research questions, report writing) will be provided. Students will have all the material from the theory section. However, students will be encouraged to carry out a literature review to get familiar with how the scientific community has approached similar cases. Teaching methods in Unit 2 are largely focused on guidance with individual groups during the Q&A sessions in weeks 7 and 8. For example, some students may need help solving coding issues, whereas others may need advice extending the methods covered in Unit 1.
Practical Guide	Theory and example applications: In this unit, students will learn theoretical concepts related to advanced techniques in probability and statistics. The unit content is spread over a period of 6 weeks. In each week, one topic will be treated comprehensively, from theory to practical applications. The following topics will be covered: (w1) Extreme value theory: Power law distributions, rank ordering statistics (including tail index estimation - e.g., Hills method), Goodness of Fit diagnosis (Kolmogorov-Smirnov, Anderson-Darling etc.), Heavy tailed distributions; Review of extreme value theory peak over threshold, block maxima; Non-stationary extreme value theory. Example assignment: data analyses of (extreme) observations for inferring, e.g., design values. (w2) Data Assimilation: maximum a posteriori probability methods, ensemble methods, variational methods and particle methods. Example assignment: quantification of subsidence due to fluid extractions. (w3) Expert Judgment: Cooks method for uncertainty quantification, Elicitation of probabilistic dependence (multivariate uncertainty). Example assignment: Elicitation of future hydro-meteorological variables due to climate change, or, Assessment of uncertainty regarding mortality and morbidity due to PM10 and PM2.5 through Cookes method. (w4) Bivariate distribution using copulas: Introduction to copulas compared to traditional bivariate distribution; Covariance, product moment correlation coefficient, rank correlation coefficients (Spearman's rho, Kendall's tau), tail dependence coefficient, estimation (maximum likelihood, moments); One parameter copulas: Gaussian copula, Gumbel copula, Frank copula, rotations of copulas. Examples of 2 parameter copulas (t-copula), and 3 parameter copulas goodness of fit diagnostic tools: semi correlations. Example assignment: modelling correlation between PM2.5 (air pollutant) and health effects (e.g., daily mortality) in the context of regression analysis for long and short term effects. (w5) Multivariate probability via Bayesian Networks: Advanced topics on discrete Bayesian networks; Non-Parametric Bayesian Networks; Dynamic Bayesian Networks. Example assignment: risk and reliability assessment, e.g., Calculation of flooding probabilities, stochastic modelling of vehicle loads in bridges, calculation of maximum bending moments in bridges, health risks, others. (w6) Multivariate probability using vine-copulas: multivariate distribution via pair-copula construction. Example assignment: multivariate probabilistic design of coastal structures subjected to multiple forces, e.g., wind speed, wave height, wave periods. Group project: Students work on a research topic within the proposed themes that they choose themselves where at least two techniques covered during the module will be applied. During week #4, the topics covered in weeks #5 and #6 will be introduced briefly so that students can better identify potential projects. The students will propose a project topic beginning in week #5, which is reviewed by the instructors and finalized by the end of week #6, including a work plan (analysis methods and anticipated results) and definition of the project data set. Weeks #7-8 include a number of Q&A sessions by the instructors for checking students progress and discuss possible common questions. While there will be some review of material, these weeks are dedicated to finalizing the group project. A project is performed in groups of 2 or 3 students (depending on the size of the class), preferably from different programs.
Assessment	Formative assessment (feedback): Formative assessments on common errors identified in weekly assignments and during projects progress meetings will be provided to the class. Group and individual formative assessments will be provided when necessary. Peer review based on weekly assignments will be implemented. This in particular for the programming components of each week. At the end of week #8 and before the final deadline for project report submission, the students will have the chance to present their partial results to their peers and receive peer feedback. The students will receive a feedback form to fill in during the group presentation. Feedback will be on the structure of the presentation, relevance of the project topic, and clarity of the methods implemented. Summative assessment: Theory and applications Summative assessment will consist of an individual oral exam covering both theoretical aspects and applications of the statistical methods taught in class. The oral exam will build upon the analyses carried out in the report and expand to the rest of the topics (theory) discussed in the module. The progress (depth and extent that each student can reach) of the oral exam will depend on the ability of each student to answer to questions. The depth and extent that each student can reach during the oral exam will determine the individual grade. The oral examination is 50% of the total grade. As stated earlier, this grade will be individual and hence may differ from the group project grade per student. LOs 1,2,3 will be tested. Group project Summative assessment consists of a written report summarizing project setup and results. The purpose of the group project is to LOs 2,4,5,6,7 will be tested. The group project is 50% of the final grade.
Special Information	All subgrades must be &#805;5.8 The methods are first discussed theoretically and then exemplified via applications. Specific applications will be drawn from applications related to the three programs in the faculty. The applications proposed will be related to a specific track in such a way that also students that do not come from that track can understand them. For example, a nonstationary statistical model of sea level height will be developed to include the signal of sea level rise. This example is relevant to coastal engineers but also to earth system scientists. At the same time, environmental engineers can understand the relevance of it without too much in-depth pre-knowledge of coastal dynamics and climate change. Students will write their own code to solve simple problems using the fundamentals of each method. In-house open-source Matlab- and python-based software will be used for more advanced problems. Example applications will be provided by members from the faculty, some of which will also be linked to the MUDE example applications and project proposals to ensure a coherent learning line in probability and statistics in the new master curriculum. During week #1 a general overview of all topics in the module will be presented, including project-related examples, to allow students time to choose the methods to implement in their final project. Starting from week #4, topics covered in weeks #5 and #6 will be introduced briefly so that students can better identify potential projects.
Expected prior Knowledge	MUDE CEGM1000

Connection to other modules	Overlap with other parts of the programmes. In the MUDE, the students will get familiar with basic concepts of probability and statistics (e.g., axioms, discrete and continuous random variables, central limit theory, bivariate distributions, Bayes theorem). These basic concepts represent a necessary pre-knowledge for understanding the topics covered in this module. We expect some overlap in extreme value analysis, and bivariate probability distributions, but this would be propaedeutic to the related more advanced topics as nonstationary extreme value analysis, and Bayesian Networks and Vine-copulas respectively. Similarly, students will be introduced to data assimilation techniques that link the data, uncertainty and modeling learning lines in the MUDE, which will also be extended in the MORE. We foresee a small overlap in HOS regarding reliability-based design approach that includes extreme value analysis. Some overlap with CO modules Data Science and Artificial Intelligence for Engineers and Monitoring of Structural Health and Geohazards mostly with respect to Bayes rule (Bayesian updating) and other data assimilation techniques is also expected.
Academic Skills	Analysis of complex problems, communicate scientific knowledge and findings in inter-disciplinary context, critical thinking, project management, team work, written and oral communication
Literature & Study Materials	Theory and applications Group project Students will receive lecture slides, references to relevant scientific literatures, text book, MOOC, recordings of past lectures on related courses, and open source software developed over the year by the instructors. All the material will be accessible via Brightspace. Example literature: Coles, Stuart, et al. An introduction to statistical modeling of extreme values. Vol. 208. London: Springer, 2001. Genest, C. and Favre, A. (2007). Everything You Always Wanted to Know about Copula Modeling but Were Afraid to Ask. J. Hydrol. Eng. 12, SPECIAL ISSUE: Copulas in Hydrology, 347368. Morales Nápoles, O., Worm, D., van den Haak, P., Hanea, A., Courage, W., Miraglia, S.: Reader for course: Introduction to Bayesian Networks, TNO-060-DTM-2013-01115, TNO, Delft, The Netherlands, 2013. Sections: 1, 2. Anca Hanea, Oswaldo Morales Napoles, Dan Ababei, Non-parametric Bayesian networks: Improving theory and reviewing applications, Reliability Engineering & System Safety, Volume 144, December 2015, Pages 265-284 MOOC material for Expert Judgment: https://online-learning.tudelft.nl/courses/decision-making-under-uncertainty-introduction-to-structured-expert-judgment/ Software: Treatment of Uncertainty: OpenTURNS, https://openturns.github.io/. Open source. Extreme value ProNEVA matlab based. Available for distribution. Discrete BNs. Netica & Hugin (commercial but a limited free version). Powerful graphical user interface Matlab & Python (open source) options less friendly graphically but easy to implement coding. Continuous BNs UniNet (commercial but a limited free version). Powerful graphical user interface BANSHEE. Matlab version available in SoftwareX (own development). Python (open source) version under development available by the end of 2021. Vine-copulas R-vine. Open source Expert Judgment ANDURIL matlab based. Available in SoftwareX (own development) ANDURYL python based. Available in SoftwareX (own development) Data Assimilation Python-based Jupyter scripts
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2006	Subsurface Storage for Energy, Water and Climate Applications	10
Responsible Instructor	Dr. H. Hajibeygi	
Module Manager	Prof.dr.ir. M. Bakker	
Module Manager	Dr. H. Hajibeygi	
Instructor	Dr. K.H.A.A. Wolf	
Instructor	Dr.ir. J.M. Bloemendal	
Instructor	Dr. B.M. van Breukelen	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Prof.dr. S. Geiger	
Contact Hours / Week x/x/x/x	Diverse: lectures, computer practicals and lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The subsurface is increasingly used to store fluids associated with energy and for climate change mitigation. There are many potential technologies, and each has its place. In this module, the key technologies which are available and under-development are detailed, including key aspects of flow of fluids in the subsurface. These concepts are utilised to evaluate the technologies, such that we can efficiently use the subsurface. Students will move from broad concepts and how to assess the applicability (Learning Unit 1), through theoretical aspects of how the systems work (Learning Unit 2) to evaluation and design (Learning Unit 3).	
Study Goals	After completion of this module the students are able: CO-SS-LO1. Explain how earth underground resources can be used and sustained in an environmentally responsible manner for cyclic storage of fluids and energy (ILO: 4, 10). CO-SS-LO2. Design a conceptual framework for a given storage project, considering the geology and geomechanics of the host rock and physics of fluids under dynamics (ILO: 1, 3, 4, 8, 11). CO-SS-LO3. Evaluate the performance of storage systems, including the influence of their design parameters and nearby systems (ILO: 4, 5, 6, 7, 9). CO-SS-LO4. Evaluate geochemical processes in order to evaluate the environmental performance of systems (ILO: 4, 5, 6, 7, 9). CO-SS-LO5. Weigh the various advantages and disadvantages of subsurface storage options against each other and propose an optimal solution (ILO: 3, 6, 9, 10). CO-SS-LO6. Communicate their findings clearly in an appropriate comprehensive manner for various technical and non-technical stakeholders (ILO: 2).	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Learning Unit 1: Subsurface storage technology - Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes. Learning Unit 2: Heat, solute and compressible fluid transport - Lectures on the transport of dissolved substances and heat in groundwater. Small scale experimental demonstrations are given during the lectures in which students participate. - Based on the scientific contents of the lectures, practical sessions will be performed to model the fundamental concepts behind successful storage of fluids in the subsurface across scales. These are mostly in the form of weekly computer modelling exercises using, for example, MODFLOW, MT3D, SEAWAT, DARTS, DARSim, or MRST. Python scripts will be used to generate input files, read output files, and analyse the results. - Group experiments in the lab will be performed to quantify physics of the stored fluids, e.g. H₂ and CO₂. Learning Unit 3: Storage design and evaluation project - Project based learning. Students will form multi-disciplinary groups. Each group will work on a specific energy scenario and conceptualise, design and evaluate a storage system based on their scenario. Groups will report findings on a weekly basis to other groups in form of oral presentations. These presentations will focus on the key messages to different stakeholders. These outputs will be formatively assessed. The final designs and evaluations will be communicated in writing (in form of report) and orally (via a presentation).	
Practical Guide	Learning Unit 1: Subsurface storage technology (1 ECTS) Link to learning objectives of the module: CO-SS: LO 1, LO 5 The various technologies related to subsurface storage and the evaluation of subsurface storage are detailed, including the current state of the art and future research directions. Theoretical and quantitative methods to gain insights into technical and environmental performance are taught. Evaluation methodologies, such as exergy analysis and multi-criteria analysis, are outlined. Topics: - Context of subsurface storage: - o global overview of Climate & Energy status - o from production to storage to mitigation. - Overview of the technologies and advancements - o Energy storage in porous media & cavities (reservoirs and salt caverns): - o Borehole thermal energy storage - o Aquifer thermal energy storage - o Hydrogen storage - o Compressed air energy storage - o Supercritical CO₂ energy storage - o Novel storage topics - o CO₂ storage - o Water storage (managed aquifer recharge (MAR)) - Assessment methods - o Exergy Analyses - o Multi-criteria analysis	

- o Environmental performance
- o Geomechanics

Learning Unit 2: Heat, solute and compressible fluid transport (6 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4

The theoretical aspects of heat, solute and compressible flow in the subsurface are presented, including the flow of heat via conduction, advection and convection, the transport of dissolved solutes and the transport of compressible fluids (e.g. CO₂ and H₂). Reactivity and the effect on transport, leading towards changes in properties or pollution are considered. The coupled behaviour of processes is detailed, e.g. the effect of temperature on physical properties and rate constants.

Topics:

- Solute transport in groundwater
- Heat transport in the surface and groundwater
- Coupled groundwater flow and concentration/heat transport
- o density and viscosity dependency on salinity and temperature
- Reactive transport of contaminants
- o Effect of temperature on rate constants.
- o Mobilization, adsorption and resorption of trace elements due to flow/concentration/temperature
- Compressible fluid flow
- o CO₂
- o H₂
- o Laboratory experiment (geochemistry): measuring CO₂ dissolution, pH change
- Subsurface flow
- o Pore scale to km scale modelling
- Simulation methods
- o Assignments and exercises using various open-source simulators: e.g. MODFLOW, MT3D, SEAWAT, PHT3D and/or PHAST, DARTS, DARSim, MRST
- Applications include seawater intrusion, managed aquifer recharge, aquifer thermal energy storage, CO₂ storage, contaminant plumes.

Learning Unit 3: Storage design and evaluation project (3 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4, LO 5, LO 6

Students are presented with an energy scenario where they must evaluate the most appropriate storage concept and present a detailed design. The design must include predictions of performance, an evaluation of environmental impact and include analysis of uncertainties. The students must demonstrate the ability to select appropriate theory, modelling concepts and data, perform numerical analysis and critically evaluate the results to form conclusions. They must clearly and precisely communicate their results to their peers.

Topics:

- Scenario evaluation
- Application of theory and tools from other two units
- Critical evaluation of results
- Presentation of project via written technical report and presentation for different stakeholders (focus on non-technical users in the final presentation)

Assessment

Formative assessment (feedback)

Learning Units 1 and 2

During the first two units weekly tasks have to be delivered. Tasks comprise written reports based on laboratory experiments, computer exercises and theoretical knowledge. In addition, class quizzes will be given.

The tasks will be formatively assessed on completeness and quality by involved lecturers. These build towards: (i) the portfolio, and (ii) the activities in Learning Unit 3.

Learning Unit 3

Groups will give weekly presentations on their progress and considering different stakeholders. The presentations will be formatively assessed on completeness and quality by involved lecturers, and peer feedback will be solicited. In addition, formative peer feedback within the group will encourage equal contribution.

Summative assessment (grading)

Portfolio of individual learning activities at the end of Learning Units 1&2

50% of final module grade.

Assessment based on:

Individual cumulative grade. Combination of (approximately) weekly assignments defined during lectures and workshops of Units 1 and 2. The students will have received formative comments after each assignment and can upgrade the assignments for the final grade.

Integrated module assessment at the end of Learning Unit 3

50% of final module grade.

Assessment based on:

Group report

Group presentation at the end of the project

Answering questions after the presentation

A 1-2 page(s) individual report on critical evaluation of all group studies (also to stimulate reading others group reports and actively ask questions during presentations and evaluate based on capacity, efficiency, safety, exergy, public perceptions,).

Both subgrades must be >= 5.8.

Expected prior Knowledge

For all MSc students CE, ENV, AES.

Academic Skills

Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.

Literature & Study Materials

Learning Unit 1: Subsurface storage technology

Books:

- o The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); and Stad x Klimaat Het gebouw als watermachine by BNA and TU Delft (2020) (in Dutch). The study is interactive available at <https://www.bna.nl/kennis/bna-onderzoek/stad-x-klimaat> which can be automatically translated by Chrome Browser to English, for international students. These books comprise and review a number of practical cases concerning integral ecoengineering solutions for socio-environmental problems occurring in the built environment.
 - o Sustainable energy without the hot air by David Mackay: provides an overview of alternative energy options.
 - o Carbon Dioxide Capture and Storage, by Intergovernmental Panel on Climate Change (UNEP): this open-access report provides guidelines for climate-driven actions related to subsurface storage of greenhouse gases, especially CO₂.
- Specific literature in form of scientific papers and book chapters on diverse subjects of subsurface storage (energy and climate) related to techno-economical, experimental, modelling, computational and analytical techniques will be provided via

	BrightSpace. Experimental material for the CO₂/H₂ storage practical, as well as for the sensor workshop will be provided during the practical. Learning Unit 2: Heat, solute and compressible fluid transport Manuals of the open-source software and the associated packages for the modelling software used. Assignments will be developed in the form of Jupyter Notebooks so that they are fully reproducible and documented. Selected papers on the modeling of ATEs, MAR and storage of compressible fluids. Learning Unit 3: Storage design and evaluation project Resources from Learning Units 1 and 2. Additional specific literature in form of scientific papers and book chapters on multi-criteria analysis and scenario analysis will be provided via BrightSpace
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2007	Resilient Deltas	10
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Instructor	Dr. J.E.A. Storms	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr. D.S. van Maren	
Instructor	Dr. A.P. Siebesma	
Instructor	Prof.dr. C.A. Katsman	
Contact Hours / Week x/x/x/x	approximately 7 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This topical cross-over deals with the societal challenge of making deltas resilient to climate change, subsidence, unprecedented economic growth and urbanisation. Future increases in sea level rise, droughts, extreme precipitation and flooding can have adverse effects on the physics and dynamics of deltas, salt intrusions and groundwater, which requires new delta interventions to be developed. The focus is on deltas globally, using the Dutch Delta as a nearby example and laboratory. The module is split in three themes: Theme 1 (3EC) - Physics of the coupled delta system under climate change and human interventions - provides students with the scientific understanding of the interaction of atmospheric, fluvial and marine processes that comprise the delta system, their response to human interventions, and their sensitivity to climate change (LO 1,2,3) Theme 2 (3EC) - Climate change and scenarios gives students hands-on experience on the use of statistical and dynamical downscaling techniques and long-term observational datasets, which they need to analyze to understand the impact of natural and man-made climate change on the delta system. Students will study and discuss different methods to construct climate scenarios, contrasting approaches developed in various countries (for example Switzerland, UK and the Netherlands) and learn how to apply them to different regions/ Focus will be on the specific climate variables: (extreme) precipitation/ droughts and sea level rise. (LO 1, 2, 4, 5) Theme 3 (4EC) - Adaptation pathways for deltas - guides students to develop pathways towards new delta technologies (flood defence and water treatment) based on climate model/data analysis using the Dynamic Adaptive Policy Pathway method among other analytical tools, and guided by SCRUM project management (LO 2, 4, 5, 6, 7, 8) Weekly Syllabus The module takes an inquiry-based approach, in which students start in week 1 by exploring a delta system of their choice and interest. They will create a conceptual map of their current understanding of that delta system including the interlinkages or feedbacks between processes (a concept map). They will do this as part of a multidisciplinary team. In week 1, one of the final deliverables of this module a proposal for an adaptation pathway for a hypothetical client/stakeholder - will be explained and discussed with the students. Student teams will serve as advisors for one delta and stakeholders for another delta. The stakeholders are responsible for climate adaptation of a delta, the advisors are hired to develop pathways. Together they will agree on a product that the advisors will deliver to the stakeholders. This product should at least contain an analysis of the local climate change effects on the main delta systems, and proposals for solutions and adaptation pathways. As examples, they might focus on flooding, salt intrusion of beach nourishments in their chosen delta system under climate change. During Weeks 28, the students will take an iterative approach towards their deliverable, enriching their conceptual understanding of the delta system in each iteration with knowledge and skills from Unit 1 & 2, which they put in practise in weekly computer labs. In the computer labs, students may develop python notebooks in which climate projected data obtained through dynamical downscaling with high resolution models are evaluated against observations. Alternatively, students can develop statistical downscaling techniques by combining global climate model data with local observations. They will also be trained in obtaining useful trends from long observational data sets of the present climate and try to attribute these to global warming. Teamwork in Unit 3 towards the deliverable will be supported with SCRUM project management. That product will be developed in three iterative cycles ("sprints"), ending in week 4, week 8 and week 10 (the final assessment).	
Study Goals	1. Explain the interaction of physical processes that take place in different components of the (coupled) delta system (ocean, rivers, land, atmosphere) 2. Explain the influence of natural variability, climate change and socio-economic development on the coupled delta system 3. Explain mechanisms and adaptation timescales associated with historic human interventions and implications for adapting to climate change 4. Construct and interpret climate change scenarios including the use of dynamical and statistical downscaling techniques based on climate model output. 5. Analyse observed and modelled trends in sea level rise, rainfall and flooding, droughts, sediment transport, subsidence and socio-economic development 6. Acquire new knowledge on green-gray solutions for mitigating impacts of human interventions and climate change, e.g. beach nourishment, river training, building with nature. 7. Develop pathways towards engineering solutions for future delta challenges using model scenarios, including uncertainties 8. Conduct a project in a multidisciplinary team	
Education Method	Unit 1: Physics of the coupled delta system under climate change and human interventions Series of lectures on fundamentals and case studies (3 hrs per week) Mini-lectures (1 hr per week) Quizzes / theory test in week 1/2 Unit 2: Climate change and scenarios Series of lectures (2 hrs per week) Computer labs (2 hrs per week) Quizzes / theory test in week 1/2 Unit 3: Adaptation pathways for deltas Discussions, supervised team work, and in-class paper reviews (2 hrs per week) Workshops in system thinking and stakeholder analysis Computer labs (2 hrs per week) Field trip to the Rhine Meuse Delta to be considered	
Literature and Study	Unit 1: Physics of the coupled delta system under climate change and human interventions	

Materials

Hoitink, A. J., Nittrouer, J. A., Passalacqua, P., Shaw, J. B., Langendoen, E. J., Huismans, Y., & van Maren, D. S. (2020). Resilience of river deltas in the Anthropocene. *Journal of Geophysical Research: Earth Surface*, 125(3), e2019JF005201.

Leuven, J. R., Pierik, H. J., Vegt, M. V. D., Bouma, T. J., & Kleinhans, M. G. (2019). Sea-level-rise-induced threats depend on the size of tide-influenced estuaries worldwide. *Nature climate change*, 9(12), 986-992.

Kirwan, M. L., Temmerman, S., Skeehean, E. E., Guntenspergen, G. R., & Fagherazzi, S. (2016). Overestimation of marsh vulnerability to sea level rise. *Nature Climate Change*, 6(3), 253-260.

Van Maren, D. S., Oost, A. P., Wang, Z. B., & Vos, P. C. (2016). The effect of land reclamations and sediment extraction on the suspended sediment concentration in the Ems Estuary. *Marine Geology*, 376, 147-157.

van Maren, D.S., J.G.W. Beemster, Z.B. Wang, Z.H. Khan, R.A Schrijvershof, A.J.F. Hoitink. Tidal amplification and river capture in response to land reclamation in the Ganges-Brahmaputra delta. *Catena*, 2022

Syvitski, J. P., Kettner, A. J., Overeem, I., Hutton, E. W., Hannon, M. T., Brakenridge, G. R., & Nicholls, R. J. (2009). Sinking deltas due to human activities. *Nature Geoscience*, 2(10), 681-686.

Syvitski, J. P., Vorosmarty, C. J., Kettner, A. J., & Green, P. (2005). Impact of humans on the flux of terrestrial sediment to the global coastal ocean. *science*, 308(5720), 376-380.

Talke, S.A. and Jay, D.A. Changing Tides: The Role of Natural and Anthropogenic Factors. *Annu. Rev.Mar. Sci.* 2020. 12:12151

Wang, Z. B., Elias, E. P., van der Spek, A. J., & Lodder, Q. J. (2018). Sediment budget and morphological development of the Dutch Wadden Sea: impact of accelerated sea-level rise and subsidence until 2100. *Netherlands Journal of Geosciences*, 97(3), 183-214.

Winterwerp JC, Wang ZB, van Braeckel A, van Holland G, Kösters F. Man-induced regime shifts in small estuariesII: a comparison of rivers. *Ocean Dyn.* 63:1293306, 2013.

Unit 2: Climate change and scenarios

van den Hurk, B. et al. Drivers of mean climate change around the Netherlands derived from CMIP5. *Clim. Dynam.* 42, 16831697 (2013).

Swiss Climate Change Scenarios CH2011 (C2SM, MeteoSwiss, ETH, NCCR Climate, and OcCC, 2011).

van den Hurk, B. et al. New climate change scenarios for the Netherlands. *Wat. Sci. Technol.* 56, 2733 (2007).

Jenkins, G. J. et al. UK Climate Projections: Briefing Report (Met Office Hadley Centre, 2009).

Frederikse, T., et al. (2020). The causes of sea-level rise since 1900. *Nature*, 584(7821), 393-397. <https://doi.org/10.1038/s41586-020-2591-3>

Tamisiea, M.E., and J.X. Mitrovica (2011). The moving boundaries of sea level change: Understanding the origins of geographic variability. *Oceanography* 24(2):2439, <https://doi.org/10.5670/oceanog.2011.25>

Slangen, A. B. A., Katsman, C. A., Van de Wal, R. S. W., Vermeersen, L. L. A., & Riva, R. E. M. (2012). Towards regional projections of twenty-first century sea-level change based on IPCC SRES scenarios. *Climate dynamics*, 38(5), 1191-1209. <https://doi.org/10.1007/s00382-011-1057-6>

Slangen, A. B. A., Carson, M., Katsman, C. A., Van de Wal, R. S. W., Köhl, A., Vermeersen, L. L. A., & Stammer, D. (2014). Projecting twenty-first century regional sea-level changes. *Climatic Change*, 124(1), 317-332. <https://link.springer.com/article/10.1007/s10584-014-1080-9>

IPCC 5AR Ch 13.1/13.6 [to be replaced by appropriate 6AR chapter this summer]

Katsman, C. A., Sterl, A., Beersma, J. J., Van den Brink, H. W., Church, J. A., Hazeleger, W., ... & Weisse, R. (2011). Exploring high-end scenarios for local sea level rise to develop flood protection strategies for a low-lying delta: the Netherlands as an example. *Climatic change*, 109(3), 617-645. <https://link.springer.com/article/10.1007/s10584-011-0037-5>

Practical Guide

Week/Unit

- Physics of Delta System / Climate
- Climate Change / Scenarios
- Adaptation pathways

1

- Deltas in the Anthropocene. An introduction to the range of systems and their response impacted by climate change and human interventions
- An introduction to climate science: climate sensitivity
- Diverse team formation, SCRUM Introduction.
 - Exploration and discussion of generic adaptation pathways and deliverables

2

- Sediment supply from rivers and from coastal seas; sediment type; changes in sediment supply (role of climate, land use & reservoirs)
- Our future climate - Representative Concentration Pathway (RCP) scenarios, climate modelling and downscaling techniques
- Selection of deltas and roles (advisor / stakeholder)
 - Systems thinking workshop & stakeholder Analysis workshop

3

- Tidal amplification response to human interventions (closure dams, channel deepening) b) Sea level rise: causes, processes + observations (altimetry / tide gauges Dutch coast)
- Discussion product deliverables and planning Sprint 1
 - Presentations Conceptual Model and Stakeholder Analysis

4

- Changes in salt intrusion and fresh water availability in response to channel deepening, reservoir construction, and channel reorganization
- Uncertainty in climate model and scenarios, climate change versus natural variability, and attribution of climate change
- Resilience debates

5

- Decline of water quality and ecological value in response to climate change and human interventions
- Atmospheric influences on deltas: Storms, surges, precipitation and river discharge
- Review and Retrospective Sprint 1 and Planning Sprint 2

6

- Subsidence and compaction of deltas
- Atmospheric influences on deltas: Projections of Extremes : Flooding and Droughts
- Discussion on evaluation of strategies

7

- Coastal erosion by natural processes, human interventions, and climate change
- Sea level rise: global and regional projections. High-end scenarios with skewed ice sheet contributions, links to extreme events / coastal safety.

8

- Morphologic response to sea level rise
- Compound effects of sea level rise, storm surges + precipitations / river discharge c) Review and Retrospective Sprint 2

	10 a) Adaptation to sea level rise c) Presentations of adaptation pathways to client/stakeholder, supported by data products consisting of e.g. downscaled climate model output for different scenarios, observational data analyses.
Assessment	Formative assessment (feedback): The formative assessment for Theme 1 and 2 is arranged in the form of quizzes and assignments with feedback. For Theme 3 the conceptual model and stakeholder analysis are formative assessed in week 3. In week 5 and 8 the SCRUM artefacts Review and Retrospective are used to give students feedback on content and process. Summative assessment: The summative assessment consists of: Written exam on content Unit 1 - 3 (50%) Proposal for an adaptation pathway / solution for a delta including analysis of the local climate change effects on the delta system, delivered in either a written/oral/(other form) presentation (50%)
Expected prior Knowledge	Students that enter the module are expected to have in-depth understanding of one of the subsystems of deltas (e.g. coastal area, rivers, the subsurface, the atmosphere etc.). All Master tracks within the programs Civil Engineering, Environmental Engineering and Applied Earth Sciences can participate. Students from other programs need to contact the module manager.
Academic Skills	The professional and personal skills are included in the learning objectives. Specifically students work on their metacognitive skills (learning objective 6) and their collaboration, management and communication skills (learning objective 8)
Literature & Study Materials	(Continuation) Unit 3: Adaptation pathways for deltas For pathway systems thinking and rapid modelling the following material (or comparable resources) will be used together will dedicated material on Brightspace: Haasnoot, M., Brown, S., Scussolini, P., Jimenez, J. A., Vafeidis, A. T., & Nicholls, R. J. (2019). Generic adaptation pathways for coastal archetypes under uncertain sea-level rise. <i>Environmental Research Communications</i>, 1(7), 071006. Wright, D., & Meadows, D. H. (2008). <i>Thinking in systems</i>. Earthscan. https://ocw.tudelft.nl/courses/ngi-2/subjects/8-bonus-module-problem-demarcation-actor-analysis-skills-2/, (accessed 2-4-2021) Goldreich, P., Mahajan, S., & Phinney, S. (1999). Order-of-magnitude physics: Understanding the world with dimensional analysis, educated guesswork, and white lies. University of Cambridge. Relevant part available via https://web.stanford.edu/class/ee353/SanjayMahajanIntro-01-1.pdf (accessed 2-4-2021) Van Berchum, E. C., van Ledden, M., Timmermans, J. S., Kwakkel, J. H., & Jonkman, S. N. (2020). Rapid flood risk screening model for compound flood events in Beira, Mozambique. <i>Natural Hazards and Earth System Sciences</i>, 20(10), 2633-2646. Jafino, B. A., Kwakkel, J. H., & Taebi, B. (2021). Enabling assessment of distributive justice through models for climate change planning: A review of recent advances and a research agenda. <i>Wiley Interdisciplinary Reviews: Climate Change</i>, e721. For teamwork and project process the following material (or comparable resources) will be used together will dedicated material on Brightspace: · Schwartz, S. H. (2014). National culture as value orientations: Consequences of value differences and cultural distance. In <i>Handbook of the Economics of Art and Culture</i> (Vol. 2, pp. 547-586). Elsevier. · https://www.mbtionline.com/ · https://scrumguides.org/ (All accessed 2-4-2021)
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2008	Monitoring of Structural Health and Geohazards	10
Responsible Instructor	Dr. E. Lourens	
Module Manager	Dr. E. Lourens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. M.J.M.M. Steenbergen	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Ir. F.J. van Leijen	
Instructor	Dr. Y. Yang	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	Dr. A. Barnhoorn	
Instructor	Dr. K. Anupam	
Instructor	C. Shen	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	12-14 hours per week (3-4 lectures, 3-4 workshops of 2*45 min)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The cross-over will be of the methodological type, oriented around processes and techniques that are shared across multiple domains. Specifically, students from CE and AES will apply predominantly new methods related to the monitoring of civil structures, rock- and soil masses to problems originating in their own as well as other fields of study. The module consists of two units: 1. Theory & Techniques 6 EC 2. Case Studies: Advanced monitoring of structural health and geohazards 4 EC Unit 1: Theory & Techniques [LO1, LO2, LO3, LO4, LO5] The first unit provides the students with knowledge on the theory and techniques required to answer the questions what, when, how, why and where to monitor?. Parts of this unit will form natural extensions to what is taught in the MUDE, including brief recapitulations. Weekly syllabus Week 1: Introduction Degradation & failure mechanisms Relation loading &#8596; materials &#8596; mechanisms &#8596; time- & spatial scale Sensors and data acquisition Advanced signal processing (beyond MUDE) Week 2: Inversion Inversion and quality assessment (including recap MUDE) Prediction (Best Linear Unbiased Prediction, Kriging, Gaussian process Regression) Inversion for sub-surface (layered) structures Week 3: System identification and model updating System identification: EMA/OMA Deterministic (FE) model updating Bayesian inference in the context of structural health and geohazards monitoring (Bayesian inversion) Week 4: Machine learning perspectives Feature engineering Introduction to image segmentation, deep learning and convolutional neural networks (including recap MUDE) Anomaly detection: clustering techniques extended from MUDE, hypothesis testing (time series analysis) Week 5: Exam Written exam (30%) Unit 2: Case studies: Advanced monitoring of structural health and geohazards [LO3, LO4, LO5, LO6] Case studies covering a wide range of application domains will be presented, where a division is made between CE applications (bridge, railway, pavement, and wind turbine monitoring, stability of dykes and embankments), AES applications (soil mass movements and landslides, land subsidence, failure & fault reactivation due to gas extraction, short-term cyclic storage of fluids or resource mining), and applications relating to both fields of study (imaging of structures, rocks and the near surface, monitoring of acoustic emission and seismic activities). Each case study will be developed & supervised by the relevant, contributing research group. Students will work in groups of 2 on these case studies, 1 from CE and 1 from AES. Each case study comprises 2 ECTS. Insofar possible, each group will have representatives from both MSc programmes. The last 4 weeks of the module are reserved solely for the case studies. Weekly syllabus Weeks 6-10: Introduction of the case studies If required: theory specific to the case study & monitoring approach used State-of-the-art review per case study	
Study Goals	[LO1] Identify mechanisms of failure and functionality loss for applications in the geo- and civil engineering domain. [LO2] Discern techniques to inspect and measure the health or integrity of civil structures, rocks, and soil masses. [LO3] Design monitoring systems to detect patterns and anomalies in civil structures, rocks and soil masses and assess the value of information obtained with these systems. [LO4] Post-process monitoring data to extract structural health information. [LO5] Apply numerical and statistical inverse analysis to identify unknowns and update numerical and/or analytical models based on the monitoring data.	

Education Method	[LO6] Integrate monitoring data and physical modelling to support decision-making. Unit 1: The content in unit 1 will be delivered through theoretical teaching sessions, flipped classrooms, live demonstrations in the classroom/laboratory, and computer workshops. Unit 2: For the case studies, the students will be split into sets of groups, where the aim is to have multiple groups working on the same case study, and directly supervised by the relevant research group. For each case study, there will first be theoretical teaching sessions to introduce the case study and point the students towards literature/articles relevant to the case study. This will be followed by regular, supervised practical sessions (per case study), where the groups can work on their assignment and ask questions.
Assessment	Formative assessment (feedback): Unit 1: Small group assignments related to the computer workshops. Unit 2: Short presentations to expert panels, partly peer-graded. During the presentations, societal aspects related to assessment decision-making should also be addressed. Summative assessment: 1. Written exam in week 5 (after Unit 1), before students start the case studies (30%) 2. Assignment portfolio based on group projects (70%) - Case study 1 (35%) - Case study 2 (35%)
Special Information	Case study example 1 [CE]: Structural health monitoring of bridges Students receive a data set containing accelerations and strains measured on a bridge. They are asked to 1) perform operational modal analysis to identify the dynamic properties of the structure, 2) update a (given) simplified model of the bridge using the identified properties, 3) design features that can be used to monitor the global vibrational response of the bridge, 4) implement an algorithm for detecting anomalies in the response, and 5) suggest improvements to the monitoring system (locations/types of sensors, data acquisition). The last task is included for all case studies where it is not practically feasible for students to collect the data themselves. Case study example 2 [AES]: Monitoring of changes in subsurface structures Students receive seismic reflection data, a subsurface fault model, and seismicity data from a subsurface reservoir (e.g. from Groningen) and are asked to 1) perform an assessment of the seismicity locations, 2) perform fault slip processes modelling, 3) assess what the role of subsurface properties has on the outcomes of these analyses, including the role of possible changes in production during continuous production, and 4) devise an improved monitoring plan to improve the quantification of subsurface seismicity properties.
Expected prior Knowledge	N/A
Connection to other modules	Overlap with other parts of the MSc programmes: CEGM1000: Data acquisition, signal processing, Bayesian inference & time series analysis is taught in the MUDE as well. For all three these topics, the theory will be briefly refreshed, and contextualized for monitoring applications. CIEM5110 Structural Mechanics and Dynamics: Also contains content on data acquisition, signal processing, and experimental modal analysis. AESB403B Induced Seismicity: Possible overlap with case studies (Groningen data may be used in this B module as well. The focus is very different, however. AESB404B Raw Material Exploration and Data Acquisition Techniques: Contains one lecture on the seismic exploration method with active and passive sources.
Academic Skills	Project management, teamwork, written and oral communication, information collection and evaluation.
Literature & Study Materials	The complete list for unit 1 still has to be compiled. For the case studies, relevant readings for the specific case study will be collected.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Earth Sciences

AES Thesis

AESM6000	Thesis Preparation	5
Responsible Instructor	Dr. A. Barnhoorn	
Responsible Instructor	Dr. M.W.N. Buxton	
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Responsible Instructor	Dr. A.A. Nuijens	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Students write a proposal for their master thesis project, which must be linked to at least one of the AES disciplines. The proposal (template will be provided) contains a review of relevant literature and resources, the research questions/objectives, hypotheses and methodology including data and software plan, a risk assessment, an ethical reflection, and a project planning. The module consists of: Learning about research, writing and communication skills and how to execute the different components of a thesis research during multiple (online) sessions. ~15% Performing a review of relevant literature and other resources to describe the state-of-the-art in the proposed thesis topic and identify gaps to be addressed in the thesis; Define the scope of the thesis, specifically what is in scope and what is out of scope (and therefore will not be addressed in the thesis); ~50% Formulating the research questions, hypotheses, and research plan. ~15% Propose an initial project planning (with Gantt-chart) that shows the sequential tasks/actions and times allocated for each task that is planned to be completed. Anticipate on possible risks and how these will be dealt with to prevent delays. Define the milestones in your planning (dates for the midterm, green light meeting and final presentation/defence) and plan regular reviews with your supervisor for feedback; ~5% Write a reflection on the societal relevance and ethical implications of the research project, based on the AES Ethics learning line of (your) research project. ~10% Perform a constructive peer review in week 8 on the thesis proposal of 2 fellow students, using the relevant assessment criteria of the AES MSc thesis proposal rubric. Students are grouped thematically, and within their group are assigned randomly to review 2 peers. ~5% Oral presentation (10 minutes + discussion) in week 10 to peers and supervisors. Presentations in thematic sessions of 4 students. Peers and supervisors give feedback based on provided criteria. Thesis preparation Final Review Meeting in week 10, which serves as the kick-off of the Thesis project in case of a Pass. The thesis proposal should provide the insight into what exactly will be studied and why, how the research is theoretically founded, how the research procedure and (data) analyses will be carried out, the timetable and the potential risks and mitigation options of the study. Note about planning: The thesis project has a nominal duration of 20 weeks. If the project is planned over a period longer than 20 weeks, there should be valid reasons* to be approved by the supervisor and this should be visible in the planning; the actual time to be spent on the thesis project is fixed to 20 weeks (i.e., 840h). If there are delays due to unforeseen circumstances* during the execution of the thesis project, the planning can still be adjusted in consultation with the supervisor(s). * Examples of valid reasons: illness, family circumstances, not being able to work full-time (job, extracurricular activities, top sports) Note about graduation with a company: The thesis proposal is to be developed in consultation with the direct supervisor(s) at TU Delft. If the thesis project takes place at a company, the graduation agreement of TU Delft must be signed by student and company. The agreement cannot have a start date before the official start of the thesis project. The company supervisor may only take an advisory role in the thesis preparation.	
Study Goals	After completing this module, students will be able to: Explore the background of the research problem by critically reviewing relevant scientific resources; Evaluate relevant theories and linking these theories to a relevant scientific problem; [ILO E] Formulate research questions and hypotheses relevant to an AES discipline which assure adequate delineation and definition of the research topic [ILO E] Identify appropriate research methodologies and techniques for collecting and analysing data needed to answer the research questions [ILO E] Reflect on the societal relevance and ethical implications and/or issues of the proposed research. [ILO F] Draft a work plan, specifying order of and links between research activities over time, to execute the research approach [ILO H]	

	Write a structured and consistent thesis proposal. [ILO G] Perform a constructive peer review on the thesis proposal of a peer-student following provided criteria. Orally present the thesis proposal in a clear and convincing manner to supervisors and peers. [ILO G] Link to intended learning outcomes programme ILO E. Formulate a research question, perform a literature, and research study, and build on existing technologies from different disciplines needed for an Applied Earth Sciences solution. ILO F. Challenge existing knowledge, show a critical attitude, and produce creative, constructive and novel solutions, and exercise independent judgement and uphold ethical standards. ILO G. Use written and oral communication skills to effectively exchange results and opinions with researchers, engineers, and Applied Earth Sciences stakeholders. ILO H. Set up, plan and monitor a project, dealing with a deadline and requirements set by Applied Earth sciences stakeholders.
Education Method	fter a collective kick-off, it will be mainly online instruction materials and videos that support students to learn about writing a research proposal and doing (thesis) research.
Assessment	Independent study/work with regular progress meetings with supervisors and peer-feedback sessions in thematic groups. Assessment (Formative) Peer review of draft proposal (submission in week 8), Feedback from supervisor on draft proposal, Presentation session with peer-review in week 10. Assessment (summative) Thesis Proposal - Report: The assessment of the module will be a pass or fail, and this will be determined by the assessment of the thesis proposal. This thesis proposal will be evaluated following assessment criteria defined in an AES MSc thesis proposal rubric. To pass this module and get access to the Master Thesis Project all items in the rubric need to be graded as sufficient (pass). The assessment committee will at least consist of one supervisor with UTQ or equivalent, and one additional staff member from the AES programme.
Expected prior Knowledge	Must have completed all first year modules, and at most 5 EC deficit. Thesis topic and supervisor to be approved before or at the start of the thesis preparation.
Academic Skills	.
Literature & Study Materials	Scientific literature and resources: Discuss with supervisor; students select some texts or other resources they will work with for their thesis project themselves within the literature review tasks. Students will (learn how to) use scholarly and library websites and databases. Comparing search results for these sites/databases is part of the discussions. Scientific writing Brightspace module: TU Write Relevant books: Cambridge academic English (2012) by Martin Hewings and Craig Thaine (ISBN 978-0-521-16521-1); The elements of Style by W. Strunk Jr. And E.B. White (available online); Eloquent Science (2009) by D.M. Schultz (ISBN 978-1-878220-91-2) Chapters on Interdisciplinary Research and Research Skills. Authors: Koen van der Gaast, Machiel Keestra, Laura Koenders, Steph Menken, Ger Post, https://ebookcentral-proquest-com.tudelft.idm.oclc.org/lib/delft/detail.action?docID=6468339
Judgement	na
Permitted Materials during Exam	n/a
Collegerama	No

AESM7000	Master Thesis Applied Earth Sciences	30
Responsible Instructor	Dr.ir. A.A. Verhagen	
Contact Hours / Week x/x/x/x	n/a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The thesis project is an individual research project based on the thesis proposal to be approved as part of the Thesis Preparation module. The thesis project has a nominal duration of 20 weeks. The actual time to be spent on the thesis project is 840 hours. If students do not work full-time, this should be reflected in the planning. Unforeseen circumstances* leading to significant delays during the execution of the project may lead to an extension to be approved by the supervisor and/or graduation coordinator. * Examples of valid reasons: illness, family circumstances, unavailability of supervisor(s), specific problems with research execution that are beyond the control of the student.	
Study Goals	After completing this module, students will be able to: Refine research questions and hypotheses relevant to Applied Earth Sciences. [ILO E] Explore the research problem by critically reviewing scientific literature and other resources; Evaluate relevant theories and apply these theories to a relevant scientific problem; Assure adequate delineation and definition of the research topic. [ILO A-E] Conduct research by selecting and applying appropriate research methodologies and techniques, collecting, and analysing data. [ILO A-F] Formulate well-founded conclusions and recommendations based on a comprehensive discussion of and critical reflection on the results, societal relevance and ethical implications. [ILO D, F, G] Execute the project as an individual and show initiative and independency in terms of research and asking for feedback. [ILO H] Execute the project using the project plan as defined in the Thesis Preparation, and adjust the plan as needed. Orally present and defend the thesis in a clear, systematic and convincing manner to a variety of audiences, including researchers, engineers, peers, and Applied Earth Science stakeholders [ILO G] Write a structured and consistent research report. [ILO G] Link to intended learning outcomes programme: ILO A. Observe, characterise and explain Earth system processes. ILO B. Develop and apply data processing and analysis techniques to analyse Earth system processes. ILO C. Model and predict Earth system processes and their variability and assess the influence of natural and anthropogenic factors. ILO D. Develop novel engineering solutions to facilitate the exploitation and/or management of Earth's natural resources in a responsible and sustainable way. ILO E. Formulate a research question, perform a literature, and research study, and build on existing technologies from different disciplines needed for an Applied Earth Sciences solution. ILO F. Challenge existing knowledge, show a critical attitude, and produce creative, constructive and novel solutions, and exercise independent judgement and uphold ethical standards. ILO G. Use written and oral communication skills to effectively exchange results and opinions with researchers, engineers, and Applied Earth Sciences stakeholders. ILO H. Set up, plan and monitor a project, dealing with a deadline and requirements set by Applied Earth sciences stakeholders.	
Education Method	The student conducts the research independently and is responsible for regularly discussing progress and results with his/her supervisors. A mid-term meeting with at least 2 TUD members of the assessment committee is planned halfway the working period. This date has to be included in the planning of the Thesis Preparation. The graduation coordinator will monitor the overall progress. If graduation takes place at a company, the student and company must have signed the TU Delft graduation agreement (contracts signed which are not in line with the graduation agreement are not valid and therefore students cannot proceed with the graduation at the company).	

Assessment	Assessment (Formative) Regular progress meetings with supervisor(s) and the student. Mid-term review halfway the project with assessment committee: student gives a presentation of progress and intermediate results. Pre-advice whether the student is going in the direction of a pass (evaluation form). Draft thesis review meeting (green-light meeting): student must submit draft of complete thesis to the complete assessment committee at least 1 week before the meeting. During the meeting, the student will give a short presentation (~20 minutes), followed by discussion and defence (will be used for final assessment). Based on the assessment of the draft thesis and performance in green-light meeting, it is decided whether the date of the final presentation and defence can be confirmed. The student receives feedback based on the assessment criteria of the MSc thesis rubric, and can use this to improve the draft. The student can start the administrative procedure to schedule his/her graduation after 20 working days. The final thesis must be provided to the committee at least 5 working days before the final presentation and defense. If the quality is deemed insufficient, the student submits a plan how to repair this in consultation with the supervisor(s), and schedules a new review meeting in a timeframe that should allow for a sufficient repair. An initial fail will be accounted for in the final grade. Final presentation and defense: students gives a presentation (~30 minutes) on the graduation project in public, followed by a Q&A by the audience, including committee. The final presentation can only take place after having successfully completed all other obligations/modules of the AES programme. Assessment (summative) The thesis project will be assessed based on the assessment criteria of the AES MSc thesis rubric. The main criteria are: Scientific approach (25%) Quality of the result/product (25%) Behavioural competences (20%) Quality of written presentation (15%) Quality of oral presentation and defence (15%) At most one partial grade for these main criteria may be lower than 5.8, but not less than 5.0. Assessment committee minimally comprises: Direct supervisor (chair), Second supervisor/assessor from TU Delft, At least two members should have UTQ or equivalent or be in process of obtaining it, and the members should be from at least 2 different sections/research groups. If graduation takes place at a company, at most one company supervisor can be a member of the assessment committee. The committee may have additional advisory members; they are not involved in the determination of the final grade.
Expected prior Knowledge	Prerequisite: Passed the Thesis Preparation.
Academic Skills	.
Literature & Study Materials	Relevant literature and other resources will be collected by the student from peer-reviewed journals, books, proceedings, internal reports etc.
Judgement	n/a
Permitted Materials during Exam	n/a
Collegerama	No

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